

Implied Markov transition matrices under structural price models

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This paper proposes an approach to compute the implied transition matrices from observations of market data on financial derivatives, when the price of the underlying originates from a structural model and the payoffs are received over a period of time. The structural price model involves a price formation mechanism which computes the price based on a set of Markovian inputs and constrained optimization processes. The developed inference method relies on a linear description of the derivative values in terms of occupation measures of the payoff duration. We establish closed-form expressions between occupation measures and state transitions, which then enable us to characterize implied state transition probabilities consistent with the market data on the derivative values. We develop methods to solve the optimization problem with the resulting nonlinear occupation measure equation. Numerical illustrations of the approach are presented for financial derivatives on network capacities. By applying the method to an electric network, we investigate the relation between financial transmission correct contract values and a range of implied probabilities of congestion in the network.

Keywords: Markov processes; Estimation of stochastic systems; Energy derivatives; Optimization

JEL Classification: C10, C60

1. Introduction

Market data contains information about market conditions and the pricing of risk. Learning market variables from traded derivatives has been practiced in financial markets, see e.g. (Jackwerth 1999, Jackwerth and Rubinstein 2012, Ross 2015). A classical example is implying volatility of the underlying price from observed option values, see e.g. (Gatheral and Taleb 2006). The methods there build on an assumed parametric model for the price dynamics of the underlying asset. The goal is thus to estimate the parameters of this price model.

With the emergence of complex and interconnected financial instruments, a contract's cash flow characteristics can depend on prices formed by a mechanism involving a variety of stochastic inputs and constrained optimization processes. Hence, an explicit expression of the derivative's value in terms of the parameters of contributing stochastic processes may not be available. Despite this challenge, computing implied processes to form the underlying price is essential to pricing such derivatives.

An example of such types of derivative contracts arises in electricity markets, namely, network congestion contracts, also called financial transmission rights (FTRs) (Hogan 1992, Rosellón and Kristiansen 2013). An electricity derivative of this type pays the holder a function of the price difference between two nodes of the transmission network, over a given period of time. The nodes correspond to injection points of the power grid. Typical holders of FTRs include utility companies or large consumers who buy electricity from remote generators (e.g. motivated by lower-cost or lower-emission considerations). Holding FTRs hedges against the risk of price variations due to network congestion (Sarkar and Khaparde 2008). To illustrate, suppose a utility company purchases its electricity from a network node i with cheaper generation capabilities and sells it at a node j . The hourly electricity prices at all nodes in the network including the nodes i and j are set by a welfare-maximizing price formation mechanism subject to several constraints, including network capacity constraints. When the capacity of the transmission network is sufficient, the prices are similar at every node and reflect the cheaper generation costs. When the network is congested however, the prices vary by node more dramatically. In

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particular, the price at j will reflect the local generation costs at j because now the generators at node i cannot provide for node j due to the binding network capacity constraints. While the utility company pays the price at node j to supply at node j , as the holder of FTRs from i to j it will receive the price difference between the nodes j and i . Hence, by holding FTRs, the total cost incurred by the utility company for supplying electricity at node j reduces to the price at node i , independently of network congestion. In addition to physical entities such as utilities, financial entities also engage in trading and repackaging FTRs (Sarkar and Khaparde 2008). Further details and discussions can be found in Kristiansen (2004), Rosellón and Kristiansen (2013), Independent Market Monitor for PJM (2018) and Harvey (2020).

The price differences forming the underlying of the FTR contracts are based on nodal prices, computed by a large-scale constrained optimization scheme (Schweppe *et al.* 1988, Ott 2003). The details of this price formation mechanism vary among different electricity markets, but its common inputs include generation offers, nodal demands, and network topology, see e.g. (Kirschen and Strbac 2018). Due to this constrained optimization scheme, the resulting prices do not admit closed-form formulas in terms of the inputs. To accurately capture the fundamental factors in the price model, structural models (Reiss and Wolak 2007) are preferred to reduced-form models such as jump diffusion or stochastic volatility models. For an overview of reduced-form electricity price models, see, e.g. Chapter 4 of Eydeland and Wolyniec (2002) or Swindle (2015). Structural models aim to stay close to the detailed price formation mechanism while providing adequate tractability for the needs of fast-paced trading and hedging. Examples of structural models for electricity prices can be found in Kanamura and Ohashi (2007), Carmona *et al.* (2013), Carmona and Coulon (2014), Moazeni *et al.* (2016), Alasseur and Feron (2018) and Ward *et al.* (2019).

More generally, structural models explicitly include some structure of the system and are typically derived based on optimizing behavior of agents while respecting the constraints governing the system under consideration (Reiss and Wolak 2007). As a result, such a structural modeling approach facilitates a more flexible description of the data and provides adaptivity to changes in the economic environment, market regulations, technological developments or market participants' behavior. Structural models are well suited for modeling commodity prices such as electricity prices, as they can explicitly incorporate considerations such as network constraints or various supply types that can directly impact the prices. In finance, structural models have been proposed to value credit default swaps (Gökgöz *et al.* 2014) and to describe systemic risk in the banking sector (Aldasoro *et al.* 2017, Kreis and Leisen 2018). Structural models have also been proposed in behavioral economics (Moore and Viscusi 1990, DellaVigna 2018) and marketing (Chintagunta *et al.* 2006).

In this paper, we consider structural models for underlying prices that are computed by a price formation mechanism based on optimization processes and Markovian stochastic inputs. The Markov processes explaining the inputs of the price formation mechanism are characterized by state transition probabilities in discrete-time or state transition rates in continuous-time. When the underlying price is expressed by a

structural model, it is nontrivial to establish an explicit relationship between the value of the derivative and the parameters of the structural model's input Markov processes. This paper proposes an inference framework for estimating the transition probabilities of the structural model's input Markov processes as implied by market data. The situation of interest is when the payoffs are received over a continuous or discrete subset of time.

The approach developed in this paper builds on a decoupled description of the contract value in terms of the payoff and expected discounted occupation measure for each state. The occupation measure is defined based on the contract's periods when payoffs are received and quantifies the cumulated time that the Markov process is expected to be in a given state, counting only the time when payoffs are received with discounting taken into account. For derivatives such as a European call option where the derivative payoff can only be received at a single time, the expected occupation measure reduces to the risk-neutral probability measure of the underlying price at maturity. For derivatives where the payoff can be received over a time set which is not a singleton, the reformulation decomposes the derivative value into contributions from each state level, formed by the payoff of the state times the discounted total time in the time set that the process is expected to remain in that state. The first component reflects the payoff function of the derivative. The latter component constitutes the occupation measure, which captures the information about the discount rate, the time set of the contract and the dynamics of the process.

A computational benefit of decoupling the payoff and the occupation measure appears when the evaluation of the payoff in a state is computationally intensive. This is the case with structural models, and in particular for FTRs, since the payoff at a state is based on electricity prices obtained from the numerical solution of a large-scale optimization problem. A change in the Markov process parameters for the purpose of sensitivity analysis or calibration will modify the occupation measure while preserving the calculated payoffs.

We show that expected occupation measures can be implied from the market data. State transition probabilities are then derived from the estimated occupation measures. As for any inverse estimation problem, this problem may be underdetermined. Therefore we identify a set of probabilities compatible with the data but do not uniquely identify probabilities. Our framework can accommodate the introduction of additional structural assumptions to further restrict the set of transition probabilities.

A key step in our inference approach is the establishment of explicit expressions between occupation measures and state transition probabilities or rates. These equations relate an inferred occupation measure to a family of transition probabilities or rates able to match this measure. The relation between the transition probabilities or rates and the occupation measure is in general nonlinear. These equations are incorporated to the estimation problem to ensure that the implied transition probabilities or rates are consistent with the market data for the contract values. The nonlinearity of these equations makes the resulting estimation problem challenging to solve. We employ a numerical optimization technique based on sequential linearization to solve this problem.

We establish closed-form representations for particular transition matrices by solving the corresponding nonlinear equations. These particular solutions also prove the feasibility of the specific nonlinear system of equations. For occupation measures in discrete-time over a finite state space, we find a particular transition matrix which is rank-one. For occupation measures in continuous-time and over a finite state space, we find a ray of transition rate matrices, where each matrix is the sum of a diagonal matrix with identical elements plus a rank-one matrix. In both cases the matrix entries are linear combinations of the initial state probabilities and expected discounted occupation measure of each state, where the weights depend nonlinearly on the discount and period duration characterizing the measure.

While the framework in this paper is general, we investigate its application to compute the probabilities of network congestion implied from the observation of financial transmission actual contract values, which can then be used to price other contracts. Using a standard benchmark network, we assess the information that can be extracted from the market data on these contracts by a parametric study. We analyze the set of implied probabilities for the specific events of interest.

Organization: The remainder of the paper is organized as follows. Section 2 is on related work. Section 3 presents the contract model. Section 4 develops the expression of contract values in terms of expected discounted occupation measures. Section 5 formulates the problem of estimating occupation measures from the observations of contract values. Section 6 relates a prescribed occupation measure to the set of Markov chains that achieve it. Section 7 describes an optimization method to handle the occupation measure constrained formulation of interest. Section 8 considers inverse estimation problems using financial transmission rights as an example of their application, and Section 9 concludes. The proofs of the propositions are provided in appendix 1.

Notation: Throughout this paper, $1_n \in \mathbb{R}^n$ denotes the column vector of ones, and $0_n \in \mathbb{R}^n$ denotes the column vector of zeros. The k th unit column vector in $\mathbb{R}^n$ is denoted by e_k . The identity matrix is denoted by I . We write $\exp(A) = \sum_{k=0}^{\infty} A^k/k!$ for the matrix exponential of a matrix A . Vectors that describe probability distributions are row vectors. We write $\leq, \geq$ for componentwise inequalities. We denote by $\mathcal{P}^n = \{p \in \mathbb{R}^{1 \times n} : \sum_{j=1}^n p_j = 1, p \geq 0\}$ the space of probability distributions over the states, $\mathcal{P}^{n \times n} = \{P \in \mathbb{R}^{n \times n} : P1_n = 1_n, P_{ij} \geq 0\}$ for the space of discrete-time transition matrices of interest, and $\mathcal{Q}^{n \times n} = \{Q \in \mathbb{R}^{n \times n} : Q_{ij} \geq 0 \text{ for } i \neq j, Q1_n = 0_n\}$ for the space of continuous-time transition rate matrices. Throughout, $\mathbb{R}_T$ denotes $[0, T]$ and $\mathbb{N}_{T-1}$ denotes $\{0, 1, \dots, T-1\}$. We distinguish the visually similar $\mathcal{T}$ (time set) from the horizon T and A^T (matrix power) from $A^\top$ (transposition). We distinguish μ, ν (measures) from μ, ν , the row vectors associated with the corresponding measures.

2. Related literature

Expected occupation measures and related notions of local time and occupation densities are introduced in the stochastic process literature, see e.g. (Darling and

Kac 1957, Kesten 1962, Pitman 1977, Eisenbaum *et al.* 2000). Local times of Markov process are discussed in the survey by Geman and Horowitz (1980) and in the monograph by Marcus and Rosen (2006). The occupation measures used in our work differ from classical occupation measures in that they take into account the discounting and accrue only during the subset of time where payoffs can be received.

In the optimal control literature, decisions variables that appear in the dual linear programming formulation of Markov Decision Processes represent state occupation measures, see (Derman and Klein 1965, Mine and Tabata 1970, de Farias and Van Roy 2003). Similarly, state occupation measures appear in linear programming formulations of constrained Markov decision processes (Feinberg and Rothblum 2012, Lee *et al.* 2014), in convex analytic methods for Markov decision processes (Bhatt and Borkar 1996, Borkar 2002, Haskell and Jain 2015), in the perturbation theory of Markov chains (Zhang and Yin 2004). In Stockbridge (2004), the pricing of barrier options and look-back options is formulated as a linear programming problem where dual variables represent occupation measures.

In quantitative finance, path-dependent derivatives that depend on the time spent in prescribed price regions have been considered in Dassios (1995), Yor (1995), Linetsky (1999), Fusai (2000), Cai *et al.* (2010), Li *et al.* (2013), Wu and Zhou (2016) and Zhou *et al.* (2017). For instance, the payoff of a particular ‘step’ European call option has been defined as $e^{-\rho\tau}[S_T - K]^+$ where ρ is a depreciation rate, T is the maturity, S_T is the underlying price at time T , K is the strike price, H is a price threshold and $\tau = \int_0^T 1_{\{S_t \leq H\}} dt$ is the cumulated time where the underlying price is below H , that is, the occupation measure of the set $\{t : S_t \leq H\}$. This payoff is smoother than the payoff of a barrier option that would pay $[S_T - K]^+$ if $S_t \geq H$ at some time $t \in [0, T]$ and 0 otherwise. Our work differs from these occupation-time derivatives in that occupation measures are not characteristic features of the contract under study, but rather, our approach decomposes the valuation of a contract into a sum of terms involving discounted occupation measures of individual states. Nevertheless, our work may be relevant to these derivatives inasmuch as the occupation measures involved there could be approximated by the more tractable occupation measures of a Markov chain.

Markov chains (finite state space Markov processes) can be used to approximate Markov processes in a continuous state space (Kushner and Dupuis 2001). This approach is developed in finance in work such as Lo and Skindilias (2014), Cui *et al.* (2018), Cui *et al.* (2019a, 2019b) and Ding *et al.* (2020). For instance, the work of Mijatovic and Pistorius (2013) describes equity and credit derivatives of firms susceptible to default through the device of time-changed Markov processes with killing. For their type of solutions, the Markov-chain approximation techniques investigated in Cui *et al.* (2018, 2019a) lead to closed-form solutions in terms of the transition matrix of the Markov chain.

The approach to derivative pricing from this line of work raises the need for calibration methods for such models. One of the techniques we employ in our work consists in expressing expected discounted occupation measures using matrix exponentiation of a matrix decomposed by blocks. The vector that encodes the occupation measure is then extracted from

a specific block of the exponentiated matrix. This formulation is motivated by subsequent optimization problems, which appear in our case when a structured regression problem is formulated to fit the model to market data. Extracting a block matrix containing optimization variables or subjecting the block to linear constraints are linear operations which are thus amenable to convex optimization. As matrix exponentiation is a nonlinear operation, we use successive linearization approximation techniques to solve a sequence of convex problems. The scheme is guaranteed to produce a valid transition matrix at each step since the constraints on the transition matrices are linear and thus can be enforced without approximation. The only approximation is the exact value of the occupation measure vector, which is recalculated after each step using the exact matrix exponentiation expression.

Inverse estimation of Markov chain models, in discrete or continuous time, has been the topic of previous studies. The determination of the set of initial probabilities compatible with given stationary and transition probabilities is the topic of Ray and Margo (1976) and Pittenger (1977). Karr (1983) focuses on the characterization of probability measures given prescribed values of Laplace transforms. The synthesis of an optimal control policy that matches a prescribed occupation measure is discussed by Krass and Vrieze (2002).

Implying stochastic process parameters from the observation of financial derivative prices has a long history in finance. The ability to infer risk-adjusted state probabilities from option prices is discussed in Ross (2015) and the references therein. Inference methods are reviewed in Jackwerth (1999), Jackwerth and Rubinstein (2012) and Cao *et al.* (2020). Computational procedures are developed in Coleman *et al.* (1999) and Li (2013). Our problem differs in the nature of the underlying, for which the prices result from a mathematical optimization procedure, and that we seek to estimate the state transition probabilities of an underlying system rather than the parameters of a parametric stochastic process driving a price.

A foundational reference on financial transmission rights (FTRs) is Hogan (1992), who describes the transmission capacity contract and the payoffs based on nodal electricity prices calculated by optimal power flow methods. See Hogan (1997) and Joskow (2005) for an overview of historical developments. FTR products, FTR markets and implementation issues are surveyed in Kristiansen (2004), Sarkar and Khaparde (2008) and Rosellón and Kristiansen (2013), while more recent descriptions can be found in the annual reports of market monitoring entities such as (Independent Market Monitor for PJM 2018). FTR pricing methods and challenges are discussed in Parmeshwaran and Muthuraman (2009). The market efficiency of FTRs and the existence of risk premia are studied in Mount and Ju (2014).

Different modeling approaches for electricity prices with their implications in pricing derivatives have been studied, see (Weron 2009) for reduced-form models, see (Carmona and Coulon 2014) for structural models, and see (Moazeni *et al.* 2016) for pricing based on more detailed generation supply curve and an approximate market clearing. Price modeling approaches which rely on an embedded price formation mechanism are capable to adapt to changes in electricity markets and thus avoid overreliance on history, while still retaining sufficient simplicity for derivatives pricing.

While the academic literature on FTRs analyzes in depth their merits, there are fewer reports on practical FTR contract pricing methodologies. Work such as Mahringer *et al.* (2015) use stylized assumptions leading to closed-form solutions. Work such as Parmeshwaran and Muthuraman (2009) adopt a simulation-based approach. In this vein, several software vendors offer solutions based on Monte Carlo simulations of a production cost model. The production cost model outputs electricity nodal prices given a system model and a scenario for the inputs. Examples of production cost models in the industry are Promod, Plexos, GE Maps, GridView (Boyd 2016). Caveats of production cost models, namely, the sensitivity to assumptions, are discussed in Kahn (1995). It is unclear how model risk considerations are incorporated and how users should define the scenarios and their weights in the simulations, but it is easy to conduct what-if analyses.

For FTRs whose payoffs are linear in the nodal prices, the value of the FTR from i to j can also be derived from forward electricity price curves relative to nodes i and j . The construction of forward electricity curves is considered in Fleten and Lemming (2003), Benth *et al.* (2007), Caldana *et al.* (2015) and Benth and Paraschiv (2018). These methods generally estimate a seasonality component from electricity market data. In Moazeni *et al.* (2016), the forward electricity curves are derived from fuel forward curves and generation supply curves. The work in the present paper relates to an inverse approach which aims to fit model parameters compatible with observations of FTR prices.

3. The contract model

Consider a contract $(f, \mathcal{T})$, where f is an instantaneous *payoff function* and $\mathcal{T}$ is the *time set* describing the set of times where instantaneous payoffs are received. The time set $\mathcal{T}$ can be a continuous or discrete subset of time. The payoff function $f : \mathbb{R}^N \mapsto \mathbb{R}$ is determined by the values of N prices that depend on the system state. The system state belongs to a state space denoted $\mathcal{S}$. The system state follows a Markov process, where $S(t)$ denotes the state at time t . The vector of prices at time t , $\pi(t) \in \mathbb{R}^N$, is defined as

$$\pi(t) := \pi^S(S(t)). \quad (1)$$

Here, π^S is a price formation mechanism specifying prices at any state. The payoff at time $t \in \mathcal{T}$ is

$$R(t) := f(\pi(t)). \quad (2)$$

The *value* of the contract $(f, \mathcal{T})$ is the expected discounted value of the payoffs, given the information available at the valuation time:

$$\begin{aligned} V_{f, \mathcal{T}} &= \mathbb{E} \left[\int_{t \in \mathcal{T}} e^{-rt} R(t) dt \right] \quad \text{for a continuous time set } \mathcal{T}, \\ V_{f, \mathcal{T}} &= \mathbb{E} \left[\sum_{t \in \mathcal{T}} \gamma^t R(t) \right] \quad \text{for a discrete time set } \mathcal{T}, \end{aligned} \quad (3)$$

where $r > 0$ is a continuous-time discount rate and $\gamma \in (0, 1)$ is the one-period discount factor. The expectation is over the

state process. All expectations are taken under the risk-neutral measure. The initial state $S(0)$ follows an initial distribution determined at valuation time, possibly concentrated on a single state. We allow $r = 0$ and $\gamma = 1$ if the time horizon is finite ($\sup\{t \in \mathcal{T}\} < \infty$).

3.1. Example: financial transmission rights contracts

Financial transmission rights (FTRs) entitle holders to revenues or charges based on day-ahead hourly locational marginal price differences attributable to grid congestion. Consider a network with N nodes representing the transmission grid. Let π^S be the price formation mechanism implemented by the Independent System Operator (ISO) which returns day-ahead nodal electricity prices $\pi(t)$ at hour t based on inputs such as load levels at nodes, generator bids, and network structure. In our presentation, the inputs are captured by the state variable $S(t)$. An obligation-type FTR between a source node i and a sink node j entitles the holder to a revenue or charge $f(\pi(t)) = \pi_j(t) - \pi_i(t)$ at each hour $t \in \mathcal{T}$. Option-type FTRs have nonnegative payoffs only: $f(\pi(t)) = [\pi_j(t) - \pi_i(t)]^+$, where $[\cdot]^+ = \max\{\cdot, 0\}$. These contracts are often long-term and typically cover a monthly, annual, or 3-year period, for all or a subset of the hours of the day (PJM Market Monitor Staff 2017). This information is described by the time set $\mathcal{T}$.

4. Contract values using occupation measures

In this section, we present the expression of contract values in terms of expected discounted occupation measures for states and the vector of state payoffs. Let $\mathcal{S}$ be the state space of the Markov process. For a set $B \subseteq \mathcal{S}$, denote the subset of $\mathcal{T}$ where the state process is in a state in B by $\mathcal{T}_B$, that is,

$$\mathcal{T}_B := \{t \in \mathcal{T} : S(t) \in B\}. \quad (4)$$

As $S(t)$ is random, the set $\mathcal{T}_B$ is random. For a given state $s \in \mathcal{S}$, let $\mathcal{T}_s$ be a shorthand for $\mathcal{T}_{\{s\}}$. If $B = \mathcal{S}$, then $\mathcal{T}_B = \mathcal{T}$ almost surely.

For $B \subseteq \mathcal{S}$, the unnormalized expected discounted occupation measure μ is thus defined by

$$\begin{aligned} \mu(\mathcal{T}_B) &:= \mathbb{E} \left[\int_{\mathcal{T}_B} e^{-rt} dt \right] \quad \text{for } \mathcal{T} \text{ continuous,} \\ \mu(\mathcal{T}_B) &:= \mathbb{E} \left[\sum_{t \in \mathcal{T}_B} \gamma^t \right] \quad \text{for } \mathcal{T} \text{ discrete.} \end{aligned} \quad (5)$$

Here, $\mu(\mathcal{T}_B)$ represents the expected time or number of periods spent by the state process in the set B , adjusted to take discounting into account.

For the special case $B = \mathcal{S}$, and consequently $\mathcal{T}_B = \mathcal{T}$, we have $\mu(\mathcal{T}) = \mathbb{E}[\int_{\mathcal{T}} e^{-rt} dt] = \int_{\mathcal{T}} e^{-rt} dt$, for a continuous set $\mathcal{T}$, and $\mu(\mathcal{T}) = \sum_{t \in \mathcal{T}} \gamma^t$ for a discrete set $\mathcal{T}$. The quantity $\mu(\mathcal{T})$ measures the duration of the time set $\mathcal{T}$, adjusted for discounting. Table 1 lists examples of time sets $\mathcal{T}$ along with

Table 1. Time sets $\mathcal{T}$ and their discounted duration $\mu(\mathcal{T})$.

Time set $\mathcal{T}$	Discount	$\mu(\mathcal{T}, r) = \int_{\mathcal{T}} e^{-rt} dt$
$\mathbb{R}_T = [0, T]$	$r = 0$	T
$\mathbb{R}_T = [0, T]$	$r > 0$	$(1 - e^{-rT})/r$
$\mathbb{R}_+ = [0, \infty)$	$r > 0$	$1/r$
Time set $\mathcal{T}$	Discount	$\mu(\mathcal{T}, \gamma) = \sum_{t \in \mathcal{T}} \gamma^t$
$\mathbb{N}_{T-1} = \{0, 1, \dots, T-1\}$	$\gamma = 1$	T
$\mathbb{N}_{T-1} = \{0, 1, \dots, T-1\}$	$0 < \gamma < 1$	$(1 - \gamma^T)/(1 - \gamma)$
$\mathbb{N} = \{0, 1, \dots\}$	$0 < \gamma < 1$	$1/(1 - \gamma)$

their corresponding measure $\mu(\mathcal{T})$. In the sequel, if we need to clarify the discount embedded in the definition of μ , we write $\mu(\mathcal{T}_B, r)$ or $\mu(\mathcal{T}_B, \gamma)$, as in table 1.

For the case $B = \{s\}$ at a particular state $s \in \mathcal{S}$, we have

$$\begin{aligned} \mu(\mathcal{T}_s) &= \mathbb{E} \left[\int_{\mathcal{T}_s} e^{-rt} dt \right] = \mathbb{E} \left[\int_{\mathcal{T}} e^{-rt} \delta_s^S(S(t)) dt \right] \\ &= \int_{\mathcal{T}} e^{-rt} \text{Prob}(S(t) = s) dt \\ \text{or } \mu(\mathcal{T}_s) &= \sum_{t \in \mathcal{T}} \gamma^t \text{Prob}(S(t) = s), \end{aligned} \quad (6)$$

where $\delta_s^S : \mathcal{S} \mapsto \{0, 1\}$ denotes the 0-1 indicator function with values $\delta_s^S(S(t)) = 1$ if the system state $S(t)$ is s , and 0 otherwise.

The normalized expected discounted occupation measure ν is defined by

$$\nu(B) := \mu(\mathcal{T}_B)/\mu(\mathcal{T}). \quad (7)$$

For a finite state space $\mathcal{S} = \{s^1, \dots, s^n\}$ where the states s^m are indexed by m with $1 \leq m \leq n$, define the row vectors $\mu, \nu \in \mathbb{R}^{1 \times n}$ with elements

$$\begin{aligned} \mu_m &:= \mu(\mathcal{T}_{s^m}), \quad 1 \leq m \leq n, \\ \nu_m &:= \nu(\{s^m\}) = \mu(\mathcal{T}_{s^m})/\mu(\mathcal{T}) = \mu_m/\mu(\mathcal{T}). \end{aligned} \quad (8)$$

From $\sum_{m=1}^n \mu(\mathcal{T}_{s^m}) = \mu(\mathcal{T})$ and $\mu(\mathcal{T}_{s^m}) \geq 0$, we conclude that ν is in the probability simplex $\mathcal{P}^n$. That is, the normalized occupation measure is a probability measure. Let $S \sim \nu$ indicate that S is drawn according to the occupation measure ν , that is, $\text{Prob}(S = s^m) = \nu_m$.

The following proposition states that a contract value is expressible as a sum over the states of the expected payoff of being in a state, times the expected discounted duration of being in the state.

PROPOSITION 4.1 *Let μ and ν be defined as in (8) and let $x \in \mathbb{R}^n$ be the payoff vector with elements $x_m = f(\pi^S(s^m))$. Then the contract value (3) can be written as*

$$\begin{aligned} V_{f, \mathcal{T}} &= \mu(\mathcal{T}) \mathbb{E}_{S \sim \nu}[f(\pi^S(S))] = \mu(\mathcal{T}) \sum_{m=1}^n \nu_m f(\pi^S(s^m)) \\ &= \mu(\mathcal{T}) \sum_{m=1}^n \nu_m x_m. \end{aligned} \quad (9)$$

The expression (9) decouples the evaluation of $V_{f,T}$ into two components: (i) The evaluation of the payoff for each state $s^m \in \mathcal{S}$, which may involve a data lookup or a computer simulation to produce the prices $\pi^S(s^m)$, but is independent of the dynamics of the state process; (ii) The evaluation of the occupation measure, which depends on the state process dynamics, time set and discounting. Equation (9) shows that the contract value $V_{f,T}$ is linear in the corresponding occupation measure.

5. Implied occupation measures from contract values

This section formulates the inference problem of occupation measures from a finite set of observation data on contract values. Consider a collection of contracts $(f^\alpha, \mathcal{T}^\alpha)$, $\alpha \in \mathcal{A}$, where f^α maps a vector of prices $\pi(t)$ to an instantaneous payoff $R^\alpha(t) = f^\alpha(\pi(t))$, and $\mathcal{T}^\alpha$ is the set of times where payoffs are received. Suppose that there are K distinct time sets $\mathcal{T}^{(1)}, \dots, \mathcal{T}^{(K)}$ in the collection $\{\mathcal{T}^\alpha\}_{\alpha \in \mathcal{A}}$. For any $\alpha \in \mathcal{A}$, let k_α be the index such that $\mathcal{T}^{(k_\alpha)} = \mathcal{T}^\alpha$. The prices are driven by the system state $S(t)$ common to these contracts, as $\pi(t) = \pi^S(S(t))$. All payoffs are subordinated to the same system state process. As a result, contracts defined on the same time sets share a common state-occupation measure. Contracts defined on different time sets use distinct state-occupation measures that are however related through the common state process.

Proposition 4.1 implies that

$$V_{f^\alpha, \mathcal{T}^\alpha} = \mu(\mathcal{T}^\alpha) \sum_{m=1}^n v_{k_\alpha, m} x_{\alpha, m}, \quad (10)$$

where $x_{\alpha, m} := f^\alpha(\pi^S(s^m))$ is the payoff from contract α when being in state s^m , and $v_{k_\alpha, m} := \mu(\mathcal{T}_{s^m}^\alpha) / \mu(\mathcal{T}^\alpha)$ is the normalized occupation measure of state m induced by $\mathcal{T}^\alpha$ and $\{S(t)\}_{t \in \mathcal{T}^\alpha}$.

Suppose we can observe the value of these contracts, denoted by $\tilde{V}_\alpha$. Let $y_\alpha = \tilde{V}_\alpha / \mu(\mathcal{T}^\alpha)$ denote normalized observations. Since contract values are linear in normalized occupation measures $v_{k_\alpha, m}$, these measures can be estimated by a linear least-squares method.

More precisely, given the normalized contract value observations y_α , the calculated $x_{\alpha, m}$, and the calculated $\mu(\mathcal{T}^\alpha)$, we can estimate the occupation measures implied by the contract values by the constrained regression problem

$$\begin{aligned} \min_{\theta \in \Theta} \quad & \|\epsilon\| + \Omega(\theta) \quad \text{subject to} \\ \epsilon_\alpha = y_\alpha - \sum_{m=1}^n \theta_{k_\alpha, m} x_{\alpha, m} \quad & \forall \alpha \in \mathcal{A}, \end{aligned} \quad (11)$$

where $\Omega(\cdot)$ is a function used to regularize the regression problem. The computed solutions $\theta_{k_\alpha, m}$ from the optimization problem (11) represent the implied normalized occupation measures $v_{k_\alpha, m}$.

As a special case, $K = 1$, i.e. all observed contracts have an identical time set $\mathcal{T}^\alpha = \mathcal{T}$ for all $\alpha \in \mathcal{A}$, we have $v_{k_\alpha, m} =$

v_m and $\theta_{k_\alpha, m} = \theta_m$. Hence, by writing X for the matrix with elements $X_{\alpha m} = x_{\alpha, m}$ (the design matrix), and treating θ as a column vector in $\mathbb{R}^n$, the problem (11) reduces to

$$\min_{\theta \in \Theta} \|y - X\theta\| + \Omega(\theta). \quad (12)$$

The benefit of (11) or (12) is to let a decision maker adapt a stochastic model of the system state to observed data on current contract values. The implied occupation measure is used as an intermediate device to infer the probabilities governing the underlying stochastic system.

In problem (11), $\theta \in \Theta$ determines constraints on the estimated occupation measures to ensure their mutual consistency. The feasible set Θ contains all θ such that there exists some transition matrix compatible with p_0 and θ . More formally, given an occupation measure ν and initial distribution p_0 , let $\mathcal{C}_{\mathcal{T}, r}^{-1}(p_0, \nu)$ denote the set of all transition matrices Q leading to ν and let $\mathcal{C}_{\mathcal{T}, \gamma}^{-1}(p_0, \nu)$ denote the set of all transition matrices P leading to ν . Then, $\Theta = \{\theta \text{ such that } \exists Q \in \mathcal{C}_{\mathcal{T}, r}^{-1}(p_0, \theta)\}$ for continuous-time Markov chains and $\Theta = \{\theta \text{ such that } \exists P \in \mathcal{C}_{\mathcal{T}, \gamma}^{-1}(p_0, \theta)\}$ for discrete-time Markov chains. Given this description of the feasible set Θ , for continuous-time Markov chains, the problem (11) can be expressed as

$$\begin{aligned} \min_{\theta_{k_\alpha} \in \mathcal{P}^n, Q \in \mathcal{Q}^{n \times n}} \quad & \|\epsilon\| + \Omega(\theta) \\ \text{s.t.} \quad & \epsilon_\alpha = y_\alpha - \sum_{m=1}^n \theta_{k_\alpha, m} x_{\alpha, m} \quad \forall \alpha \in \mathcal{A}, \\ & Q \in \mathcal{C}_{\mathcal{T}^\alpha, r}^{-1}(p_0, \theta_{k_\alpha}), \quad \forall \alpha \in \mathcal{A}. \end{aligned} \quad (13)$$

For discrete-time Markov chains, a similar reformulation can be adopted. In the optimization problem, both the occupation measure vector $\nu \in \mathcal{P}^n$ and the transition matrices $Q \in \mathcal{Q}^{n \times n}$ or $P \in \mathcal{P}^{n \times n}$ are decision variables. In general, a solution Q from problem (13) is not unique. Structural assumptions on the transition function can reduce the number of parameters to estimate.

For the special case where $\mathcal{T}^\alpha = \mathcal{T}$ for all $\alpha \in \mathcal{A}$, one can relax the constraint $Q \in \mathcal{C}_{\mathcal{T}^\alpha, r}^{-1}(p_0, \theta)$ and solve problem (12) with $\Theta = \mathcal{P}^n$ in order to obtain the implied occupation measure corresponding to that time set. The implied occupation measure can then be employed in Proposition 4.1 to price contracts with the same time set but different payoff structures. In a separate step, one can also use the implied occupation measure to compute implied transition matrix governing the underlying stochastic system. However, for $K \geq 2$, it is essential to explicitly impose the set of constraints $Q \in \mathcal{C}_{\mathcal{T}^\alpha, r}^{-1}(p_0, \theta_{k_\alpha})$ to ensure that the implied occupation measures correspond to the same stochastic system. As a byproduct, both implied occupation measures and transition functions are obtained by solving problem (13). This implied transition function can then be used to compute occupation measures corresponding to different time sets, and consequently to value contracts with not only different payoff structures but also different time sets. In addition, the implied transition function can be used to value other types of contracts whose

prices are not available from the market, perhaps due to illiquidity. Section 6 is dedicated to characterize implied transition functions and the set $\mathcal{C}_{T,r}^{-1}(p_0, \theta)$.

6. Implied transition matrices from occupation measures

This section describes the set of transition functions that are compatible with a prescribed occupation measure given a time set, discount, and initial state distribution associated with the occupation measure. In section 6.1, we establish closed-form expressions for the occupation measures for several typical time sets. The set of transition functions compatible with a prescribed occupation measure is characterized in Section 6.2. Conditions for this set to be nonempty are given in section 6.3. Closed-form solutions to the nonlinear equations defining the set of transition functions are provided in section 6.4.

6.1. Closed-Form expression for occupation measures

Let $p(t) \in \mathcal{P}^n$ be the row vector describing the state probabilities at time t . Let $p_0 = p(0)$ be the row vector of initial state probabilities. In continuous-time, the transition function is described by a state transition rate matrix $Q \in \mathcal{Q}^{n \times n}$, and $p(t) = p(0) \exp(Qt)$. In discrete-time, the transition function is described by a state transition probability matrix $P \in \mathcal{P}^{n \times n}$, and $p(t) = p(0)P^t$ (matrix power). Throughout, (p_0, Q) denotes a continuous-time Markov chain with initial distribution $p_0 \in \mathcal{P}^n$ and transition rate matrix $Q \in \mathcal{Q}^{n \times n}$. Similarly, (p_0, P) denotes a discrete-time Markov chain with initial distribution p_0 and transition probability matrix $P \in \mathcal{P}^{n \times n}$.

Recall that $\mu \in \mathbb{R}^{1 \times n}$ has components $\mu_m = \mu(\mathcal{T}_m)$. From (6) and $p(t)$ we have

$$\mu = p_0 \int_{t \in \mathcal{T}} \exp((Q - rI)t) dt \quad \text{or} \quad \mu = p_0 \sum_{t \in \mathcal{T}} (\gamma P)^t. \quad (14)$$

Note that if a matrix $A \in \mathbb{R}^{n \times n}$ is invertible, we have $\int_0^T \exp(At) dt = (\exp(AT) - I)A^{-1}$. Otherwise, we can use a consequence of a theorem from Van Loan (1978): for any $A \in \mathbb{R}^{n \times n}$, $u \in \mathbb{R}^{1 \times n}$, it holds that $\int_0^T u \exp(At) dt = Y_{12}$ where Y_{12} is obtained by identifying $\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \exp\left(\begin{bmatrix} 0 & u \\ 0_n & A \end{bmatrix} T\right)$. Similarly, in discrete-time, if $0 < \gamma < 1$ we have $\sum_{t=0}^{T-1} (\gamma P)^t = (I - \gamma^T P^T)(I - \gamma P)^{-1}$. For $0 < \gamma \leq 1$, which includes the previous case and the case $\gamma = 1$, we have $\sum_{t=0}^{T-1} p_0 (\gamma P)^t = Y_{12}$ where $\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} 1 & p_0 \\ 0_n & \gamma P \end{bmatrix}^T$. This relation can be verified by induction on T . This leads to Proposition 6.1, which is proved in appendix 1.

PROPOSITION 6.1 *For each time set $\mathcal{T}$, discount r or γ and Markov process (p_0, P) or (p_0, Q) listed in table 2, the corresponding listed μ has the components $\mu_m = \mu(\mathcal{T}_m)$.*

These results suffice to evaluate occupation measures over arbitrary time sets, using the additivity of the unnormalized measure μ . For instance, on $\mathcal{T} = (a, b]$, one uses $\mathcal{T} = [0, b] \setminus [0, a]$ (set difference of intervals) and calculates μ as $\mu = \mu^b - \mu^a$ where μ^b is the occupation on $[0, b]$ and μ^a is the occupation on $[0, a]$, obtained from the tabulated expressions.

Table 2. Unnormalized occupation $\mu(\mathcal{T}_{sm})$ for time-homogeneous Markov processes.

$\mathcal{T}$	Discount	Process	Occupation $\mu \in \mathbb{R}^{1 \times n}$, where $\mu_m = \mu(\mathcal{T}_{sm}) = v_m \mu(\mathcal{T})$
$\mathbb{R}_T$	$r \geq 0$	(p_0, Q)	$\mu = Y_{12}$ where $\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \exp\left(\begin{bmatrix} 0 & p_0 \\ 0_n & Q - rI \end{bmatrix} T\right)$
$\mathbb{R}_T$	$r > 0$	(p_0, Q)	$\mu = p_0 [I - \exp(-[rI - Q]T)](rI - Q)^{-1}$
$\mathbb{R}_+$	$r > 0$	(p_0, Q)	$\mu = p_0 (rI - Q)^{-1}$
$\mathbb{N}_{T-1}$	$0 < \gamma \leq 1$	(p_0, P)	$\mu = Y_{12}$ where $\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} 1 & p_0 \\ 0_n & \gamma P \end{bmatrix}^T$
$\mathbb{N}_{T-1}$	$0 < \gamma < 1$	(p_0, P)	$\mu = p_0 (I - \gamma^T P^T)(I - \gamma P)^{-1}$
$\mathbb{N}$	$0 < \gamma < 1$	(p_0, P)	$\mu = p_0 (I - \gamma P)^{-1}$

The unnormalized occupation measure calculations can be extended to non-homogeneous discrete-time Markov chains as follows. Let $(p_0, \{P_t\}_{t \in \mathbb{N}})$ describe a non-homogeneous Markov chain, such that $p(t) = p_0 \prod_{k=0}^{t-1} P_k$. Let $\mathcal{T}$ be a general time set and let δ_t be the 0-1 indicator that $t \in \mathcal{T}$. Then we have $\mu = \sum_{t \in \mathcal{T}} \gamma^t p_0 \prod_{k=0}^{t-1} P_k = p_0 \sum_{t=0}^{\infty} \delta_t \prod_{k=0}^{t-1} (\gamma P_k)$ and $\mu(\mathcal{T}) = \sum_{t=0}^{\infty} \delta_t \gamma^t$. Assuming $\sup(t \in \mathcal{T}) = T - 1$, we find that

$$\mu = p_0 Z_{12} \quad \text{where} \quad \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \prod_{t=0}^{T-1} \begin{bmatrix} \gamma P_t & \delta_t I \\ 0 & I \end{bmatrix} \in \mathbb{R}^{2n \times 2n},$$

$$\mu(\mathcal{T}) = z_{12} \quad \text{where} \quad \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} = \prod_{t=0}^{T-1} \begin{bmatrix} \gamma & \delta_t \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 2}. \quad (15)$$

These relations can be verified by induction on T .

6.2. Nonlinear equations characterizing the set of transition matrices

Proposition 6.2 characterizes the set of transition matrices that can realize a prescribed normalized occupation measure vector v .

PROPOSITION 6.2 *Consider a continuous-time Markov chain (p_0, Q) with an unknown transition rate matrix $Q \in \mathcal{Q}^{n \times n}$. Let $r > 0$. For a given discounted occupation measure $v \in \mathcal{P}^n$, the set $\mathcal{C}_{T,r}^{-1}(p_0, v)$ is given by*

$$\mathcal{C}_{\mathbb{R}_T,r}^{-1}(p_0, v) = \left\{ Q \in \mathcal{Q}^{n \times n} : rp_0 \frac{(I - \exp(-(rI - Q)T))}{(1 - e^{-rT})} + vQ = rv \right\}, \quad (16)$$

$$\mathcal{C}_{\mathbb{R}_+,r}^{-1}(p_0, v) = \{Q \in \mathcal{Q}^{n \times n} : rp_0 + vQ = rv\}. \quad (17)$$

Similarly, consider a discrete-time Markov chain (p_0, P) with an unknown transition probability matrix $P \in \mathcal{P}^{n \times n}$. Let $0 <$

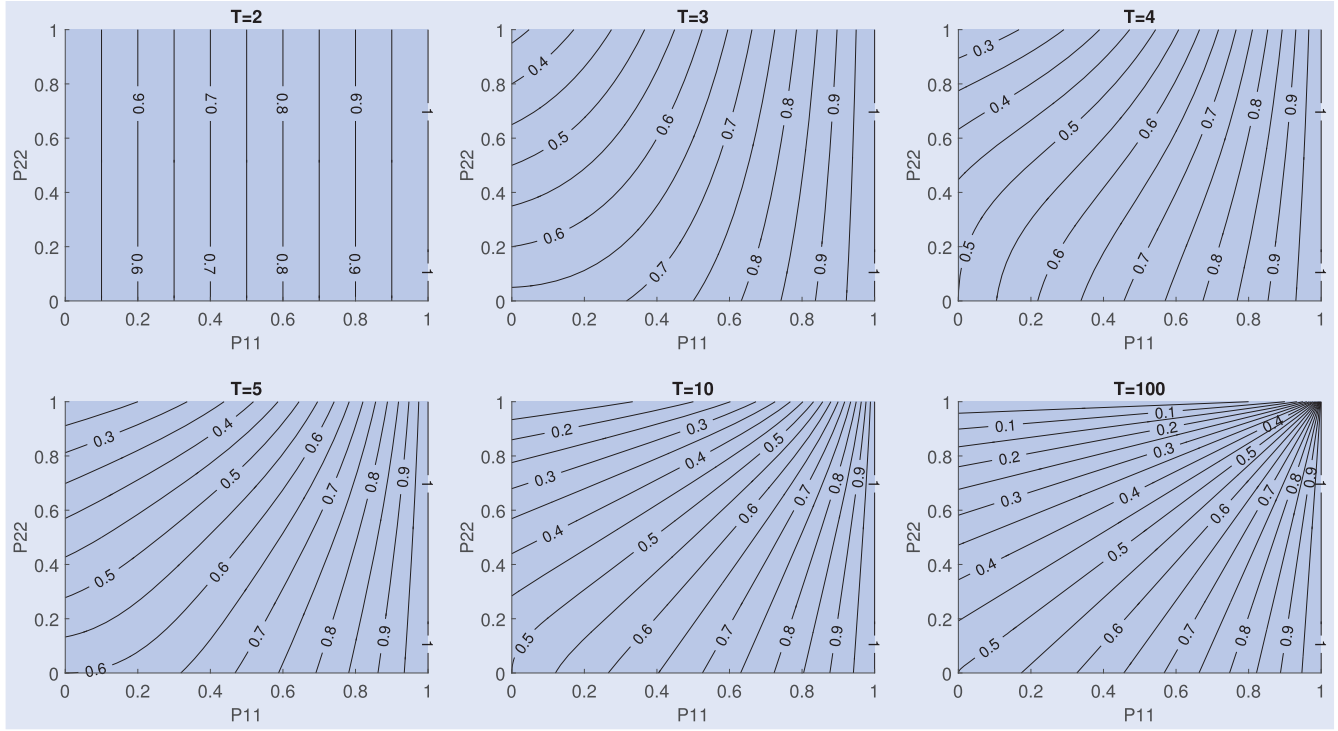


Figure 1. Sets of transition matrices $P \in \mathcal{P}^{2 \times 2}$ corresponding to a given occupation v , for various T and v . The curves are labeled by v_1 , which fully determines $v = [v_1, 1 - v_1]$. Each curve in a plane (P_{11}, P_{22}) fully describes $\mathcal{C}_{\mathbb{N}_{T-1}}^{-1}(p_0, v)$ for a particular v .

$\gamma < 1$. For a prescribed discounted occupation measure $v \in \mathcal{P}^n$, the set $\mathcal{C}_{T,\gamma}^{-1}(p_0, v)$ is given by

$$\mathcal{C}_{\mathbb{N}_{T-1},\gamma}^{-1}(p_0, v) = \{P \in \mathcal{P}^{n \times n} : (1 - \gamma)p_0 + (I - \gamma^T P^T)/(1 - \gamma^T) + \gamma v P = v\}, \quad (18)$$

$$\mathcal{C}_{\mathbb{N},\gamma}^{-1}(p_0, v) = \{P \in \mathcal{P}^{n \times n} : (1 - \gamma)p_0 + \gamma v P = v\}. \quad (19)$$

In the undiscounted cases ($r = 0$ or $\gamma = 1$), one should describe $\mathcal{C}^{-1}(p_0, v)$ by using instead $v = \mu/T = Y_{12}/T$ with Y_{12} as in table 2.

The set $\mathcal{C}^{-1}(p_0, v)$ is convex in P or Q for the two infinite-horizon cases. Indeed, the sets $\mathcal{P}^{n \times n}$ and $\mathcal{Q}^{n \times n}$ are convex polyhedral, and the infinite-horizon cases $T = \mathbb{R}_+$ and $T = \mathbb{N}$ lead to a linear equality when v is fixed. In the finite-horizon cases, the equality constraints are nonlinear due to the matrix exponential or the matrix power.

Figure 1 illustrates the case $T = \mathbb{N}_{T-1}$ for the family of 2-state homogeneous Markov chains (p_0, P) starting from the first state, with $\gamma = 1$ (undiscounted case) and the initial distribution $p_0 = [1 \ 0]$. Since $P \in \mathcal{P}^{2 \times 2}$ has only 2 degrees of freedom, each point in the plane (P_{11}, P_{22}) uniquely determines a transition matrix $P = \begin{bmatrix} P_{11} & 1 - P_{11} \\ 1 - P_{22} & P_{22} \end{bmatrix}$. The equation $v = Y_{12}/T$ for a given $v = [v_1 \ v_2]$, with $v_2 = 1 - v_1$, determines the set $\mathcal{C}_{\mathbb{N}_{T-1}}^{-1}(p_0, v)$. This set can here be identified by a curve in the (P_{11}, P_{22}) plane.

The curves of figure 1 are nonlinear. As T tends to infinity, the curves become linear. In fact, the elements of $\mathcal{C}^{-1}(p_0, v)$ would ultimately satisfy the equation $vP = v$, affine in P , obtained from the case $T = \mathbb{N}$ with $\gamma \rightarrow 1$.

It can be observed that diagonally dominant matrices need smaller adjustments to match a change in the target occupation measure. This effect is visible on figure 1: the higher concentration of ‘iso- v_1 curves’ in the upper-right quadrant compared to the lower-left quadrant means that transition matrices corresponding to the upper-right quadrant (higher P_{11} and P_{22}) need smaller adjustments to match a change in v_1 .

6.3. Domain of the inverse mapping

Given p_0 , T , and the discount rate r or discount factor γ , the domain of the set-valued mapping $v \mapsto \mathcal{C}^{-1}(p_0, v)$ is the set of occupation measure vectors v that can be achieved by some Markov chain.

Formally, using $\emptyset$ to denote the empty set, the domain is defined as

$$\text{dom } \mathcal{C}^{-1}(p_0, \cdot) = \{v \in \mathbb{R}^{1 \times n} : \mathcal{C}^{-1}(p_0, v) \neq \emptyset\}. \quad (20)$$

In continuous-time, there is no restriction on v and thus the domain is trivially $\mathcal{P}^n$, as long as we do not further restrict the domain of Q , e.g. by bounding the rates with $Q_{ij} \leq \bar{q}_{ij}$ for $i \neq j$. In discrete-time, the domain depends on the initial condition and the discount factor. For instance, figure 1 shows that with $p_0 = [1 \ 0]$, $T = 2$ and $\gamma = 1$, only $0.5 \leq v_1 \leq 1$ is achievable.

PROPOSITION 6.3 *Let γ be the discount factor.*

- (a) *For the time set $T = \mathbb{N}$, the domain of $\mathcal{C}_{\mathbb{N},\gamma}^{-1}(p_0, \cdot)$ is described by*

$$\text{dom } \mathcal{C}_{\mathbb{N},\gamma}^{-1}(p_0, \cdot) = \{v \in \mathcal{P}^n : (1 - \gamma)p_0$$

$$\leq v \leq (1 - \gamma)p_0 + \gamma 1_n^\top. \quad (21)$$

- (b) For the time set $\mathcal{T} = \mathbb{N}_{T-1}$, the domain of $\mathcal{C}_{\mathbb{N}_{T-1}, \gamma}^{-1}(p_0, \cdot)$ is given by

$$\begin{aligned} \text{dom } \mathcal{C}_{\mathbb{N}_{T-1}, \gamma}^{-1}(p_0, \cdot) &= \left\{ v \in \mathcal{P}^n : \left(\frac{1 - \gamma}{1 - \gamma^T} \right) p_0 \right. \\ &\quad \left. \leq v \leq \left(\frac{1 - \gamma}{1 - \gamma^T} \right) p_0 + \left(\frac{\gamma - \gamma^T}{1 - \gamma^T} \right) 1_n^\top \right\}. \end{aligned} \quad (22)$$

- (c) The domain of $v \mapsto \mathcal{C}^{-1}(p_0, v)$, described by (21) or (22), is convex.
- (d) The set-valued mapping $T \mapsto \text{dom } \mathcal{C}_{\mathbb{N}_{T-1}, \gamma}^{-1}(p_0, \cdot)$ is increasing in T , in the sense that $T < T'$ implies $\mathcal{C}_{\mathbb{N}_{T-1}, \gamma}^{-1}(p_0, \cdot) \subset \mathcal{C}_{\mathbb{N}_{T'-1}, \gamma}^{-1}(p_0, \cdot)$.
- (e) The domain of $v \mapsto \mathcal{C}^{-1}(p_0, v) \cap \mathcal{X}$, where $\mathcal{X}$ is a convex set restricting the set of transition matrices of interest, is a subset of $\text{dom } \mathcal{C}^{-1}(p_0, v)$, and is possibly nonconvex.

Thus, even though $\mathcal{C}^{-1}(p_0, v)$ is in general nonconvex, the domain of $v \mapsto \mathcal{C}^{-1}(p_0, v)$ is itself convex. The proof that the domain is convex, given in appendix 1, is established by constructing a closed-form solution P_v guaranteed to be in $\mathcal{C}^{-1}(p_0, v)$ whenever v is realizable by some Markov chain. Section 6.4 details those closed-form solutions.

Proposition 6.3 also states that adding constraints on P can make the domain nonconvex. For instance, if some state transitions are impossible, one can impose $P \in \mathcal{X}$ where $\mathcal{X}$ is the set of transition matrices with $P_{ij} = 0$ for some ij . In this case however, the domain of $v \mapsto \mathcal{C}^{-1}(p_0, v) \cap \mathcal{X}$ can become nonconvex. This is proven in appendix 1 by exhibiting an example.

6.4. Particular closed-form expression for implied transition matrices

The proof of Proposition 6.3 works by constructing a closed-form solution P_v guaranteed to be in $\mathcal{C}^{-1}(p_0, v)$ whenever v is realizable by some Markov chain. The following proposition (proved in appendix 1) establishes closed-form solutions for the implied transition matrices.

Below, the function $W(\cdot)$ refers to the upper branch of the Lambert-W function (see e.g. Corless *et al.* 1996).

PROPOSITION 6.4 *Let the initial condition p_0 , the time set $\mathcal{T}$, and the discount rate r or discount factor γ be fixed. Suppose that the domain of $v \mapsto \mathcal{C}^{-1}(p_0, v)$ is nonempty. Then, the following transition matrices achieve a prescribed occupation measure v .*

- (a) For $\mathcal{T} = \mathbb{R}_+$, a solution $Q_v \in \mathcal{C}_{\mathbb{R}_+, r}^{-1}(p_0, v)$ parametrized by β is

$$\begin{aligned} Q_v &= (\beta - r)[1_n b_v^\top - I], \quad b_v^\top = \frac{\beta v - r p_0}{\beta - r} \in \mathbb{R}^{1 \times n}, \\ \beta &\geq r c_0, \end{aligned} \quad (23)$$

where $c_0 = \max_{m \in \{1, \dots, n\}} \left\{ \frac{p_{0,m}}{v_m} \right\}$, with the convention $0/0 = 0$.

- (b) For $\mathcal{T} = \mathbb{R}_T$, a solution $Q_v \in \mathcal{C}_{\mathbb{R}_T, r}^{-1}(p_0, v)$ parametrized by β is

$$\begin{aligned} Q_v &= (\beta - r)[1_n b_v^\top - I], \\ b_v^\top &= \frac{\beta(1 - e^{-rT})v - r(1 - e^{-\beta T})p_0}{\beta(1 - e^{-rT}) - r(1 - e^{-\beta T})}, \end{aligned} \quad (24)$$

where

$$\begin{aligned} \beta &\geq \frac{c + W(-ce^{-c})}{T}, \quad c = \frac{rT}{1 - e^{-rT}} c_0, \\ c_0 &= \max_m \left\{ \frac{p_{0,m}}{v_m} \right\}, \end{aligned}$$

and $W(\cdot)$ is the upper branch of the Lambert-W function. For $r = 0$, we use $\lim_{r \rightarrow 0} r^{-1}(1 - e^{-rT}) = T$. In (24), the particular solution $\beta = c/T$ is always feasible. However, with the minimal solution $\beta = (c + W(-ce^{-c}))/T$, the vector $b_v^\top$ simplifies to $b_v^\top = (c_0 v - p_0)/(c_0 - 1)$.

- (c) For the time set $\mathcal{T} = \mathbb{N}$, a rank-1 solution $P_v \in \mathcal{C}_{\mathbb{N}, \gamma}^{-1}(p_0, v)$ is given by

$$P_v = 1_n d_v^\top, \quad d_v^\top = \left(\frac{1}{\gamma} \right) v - \left(\frac{1 - \gamma}{\gamma} \right) p_0 \in \mathbb{R}^{1 \times n}. \quad (25)$$

- (d) For the time set $\mathcal{T} = \mathbb{N}_{T-1}$, a rank-1 solution $P_v \in \mathcal{C}_{\mathbb{N}_{T-1}, \gamma}^{-1}(p_0, v)$ is given by

$$\begin{aligned} P_v &= 1_n d_v^\top, \\ d_v^\top &= \left(\frac{1 - \gamma^T}{\gamma - \gamma^T} \right) v - \left(\frac{1 - \gamma}{\gamma - \gamma^T} \right) p_0 \in \mathbb{R}^{1 \times n}. \end{aligned} \quad (26)$$

For $\gamma = 1$, the passage to the limit gives $d_v^\top = \frac{T}{T-1} v - \frac{1}{T-1} p_0$.

The parameter β in (23) and (24) has the dimension of a discount rate, with $\beta \geq r$ as long as the stated nonempty domain assumption holds (ensuring $1 \leq c < \infty$). The choice $\beta = r$ is feasible only when $p_0 = v$, with the solution $Q_v = 0$ corresponding to this choice of β .

The vector $b_v^\top$ in (23), (24), describes the long-term probabilities of the continuous-time Markov chain, since $b_v^\top$ verifies the steady-state condition $b_v^\top Q_v = (\beta - r)[b_v^\top 1_n b_v^\top - b_v^\top I] = (\beta - r)(b_v^\top - b_v^\top) = 0$. Similarly, $d_v^\top$ in (25), (26) describes the long-term probabilities of the discrete-time Markov chain, since $d_v^\top$ satisfies the steady-state condition $d_v^\top P_v = d_v^\top 1_n d_v^\top = d_v^\top$.

Referring back to the illustration of figure 1 for a 2-state Markov chain with $\gamma = 1$, we are in the case $P = 1 d_v^\top$ with $d_v^\top = \frac{T}{T-1} v - \frac{1}{T-1} p_0$. As $P = 1_n d_v^\top$ in dimension 2 implies $P_{12} = P_{22}$ and thus $P_{11} + P_{22} = 1$, the closed-form solutions are located at the intersection of the curves of figure 1 with the line segment connecting the points (0, 1) and (1, 0) in the (P_{11}, P_{22}) plane.

In continuous time, the parameter β allows us to describe more than one solution. In fact, the closed-form rank-1 solutions solve the inverse problem for two-state Markov chains:

PROPOSITION 6.5 *In dimension 2, $\mathcal{C}_{\mathbb{R}_T, r}^{-1}(p_0, v)$ is the set of matrices $Q_v \in \mathbb{R}^{2 \times 2}$ obtained by (24) with β ranging over the specified ray.*

6.5. Example: reliability

We illustrate the results on a simple example. Suppose a system is made of two parallel components, each of which may fail. The system can be in three states, labeled as 0 (no failure), 1 (one failure, any component), 2 (two failures). We take $T = 1$ year, and $r = 0.03/\text{year}$. The occupation measure is described by $v_0 = 0.95$, $v_1 = 0.04$, $v_2 = 0.01$.

The system starts in state 0. A transition rate matrix compatible with the data can be obtained via (24) with the minimal value of β :

$$\begin{aligned} c_0 &= 1.0526, \quad c = 1.0685, \quad \beta = 0.1340, \\ b_v^\top &= [0 \quad 0.8 \quad 0.2], \\ Q_v &= \begin{bmatrix} -0.1040 & 0.0832 & 0.0208 \\ 0.0000 & -0.0208 & 0.0208 \\ 0.0000 & 0.0832 & -0.0832 \end{bmatrix}. \end{aligned}$$

This is somewhat unsatisfying since failed components are never repaired. Thus assume now that the mean repair time is about one week: $t_r = 1/50$. We can pick a different value for β to get $Q_{10} = 1/t_r = 50$. With the value $\beta' = 52.716$ (found by increasing β until $Q_{10} = 50$) one gets

$$\begin{aligned} b_v^\top &= [0.9490 \quad 0.0408 \quad 0.0102], \\ Q'_v &= \begin{bmatrix} -2.6860 & 2.1488 & 0.5372 \\ 50.0 & -50.5372 & 0.5372 \\ 50.0 & 2.1488 & -52.1488 \end{bmatrix}. \end{aligned}$$

7. Numerical methods for occupation measure inference

This section presents methods to solve optimization problems of the form (13). The relation between the decision variables v and Q (or P) is captured by a nonlinear equality constraint, as seen in table 2. The first row in table 2 corresponding to $\mathcal{T} = \mathbb{R}_T$ implies that in continuous-time this constraint can be expressed in the form $Y = \exp(XT)$. The fourth row of table 2 corresponding to $\mathcal{T} = \mathbb{N}_{T-1}$ implies that in discrete-time this constraint is represented as $Y = X^T$. Here, X and Y are characterized by v and Q (or P). Therefore, seeking an optimal solution of problem (13) involves solving optimization problems of the following forms

$$\min_{X, Y} f(X, Y) \quad \text{such that } (X, Y) \in \mathcal{K}, \quad Y = \exp(XT), \quad (27)$$

$$\text{or } \min_{X, Y} f(X, Y) \quad \text{such that } (X, Y) \in \mathcal{K}, \quad Y = X^T, \quad (28)$$

where f is a convex function and $\mathcal{K}$ is a convex set.

We can approach these problems by solving a sequence of optimization problems with linearized constraints. This

is a frequent technique in nonlinear optimization (Gould et al. 2005, Nocedal and Wright 2006). Given an iterate Y_k , we replace $Y = \exp(XT)$ (or $Y = X^T$) by the constraint $Y = c_{k+1}(X)$, where c_{k+1} is a linear approximation of the matrix exponential (or matrix power) in a neighborhood of Y_k . The solution to the problem reformulated with the linearized constraint $Y = c_{k+1}(X)$ furnishes the next iterate Y_{k+1} . It is also possible to soften the equality constraint during the intermediate iterations by instead penalizing the difference between Y and $c_{k+1}(X)$ in the objective function. Alternatively, one can confine the update $Y - Y_k$ to a trust region.

7.1. Linearization of the matrix power

We detail the linearization of $Y = X^T$ around X_k . Using matrix differential calculus (Magnus and Neudecker 1999), $d(X^T) = d(X \dots X) = d(X)X^{T-1} + \dots + X^{T-1}d(X)$, the first-order approximation of X^T around X_k is

$$c_{k+1}(X) = X_k^T + \sum_{\ell=0}^{T-1} X_k^\ell (X - X_k) X_k^{T-1-\ell}. \quad (29)$$

This linearization is more evident by considering the linearization of the matrix product $Y = \prod_{t=0}^{T-1} X_t$, which appears in (15) to describe the occupation measure of a non-homogeneous Markov chain. In this case the linearization around $(X_{0,k}, \dots, X_{T-1,k})$ is

$$\begin{aligned} c_{k+1}(X_0, \dots, X_{T-1}) \\ = \left(\prod_{t=0}^{T-1} X_{t,k} \right) + \sum_{\ell=0}^{T-1} \left(\prod_{t=0}^{\ell-1} X_{t,k} \right) (X_\ell - X_{\ell,k}) \left(\prod_{t=\ell+1}^{T-1} X_{t,k} \right). \end{aligned}$$

If T is large, (29) is expensive to evaluate. The following proposition furnishes an alternative approach, which is particularly efficient when X (and thus X_k) have a fixed sparsity structure.

PROPOSITION 7.1 *Consider $Y = X^T \in \mathbb{R}^{d \times d}$ and its linearization c_{k+1} around X_k . Let $g_{k+1} : \mathbb{R}^{d \times d} \mapsto \mathbb{R}^{d \times d}$ be the mapping with values*

$$g_{k+1}(E) = Y_{12}, \quad \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} X_k & E \\ 0 & X_k \end{bmatrix}^T \in \mathbb{R}^{2d \times 2d}. \quad (30)$$

Then, $c_{k+1}(X) = X_k^T + \sum_{ij} (X_{ij} - X_{k,ij}) g_{k+1}(e_i e_j^\top)$.

7.2. Linearization of the matrix exponential

We detail the linearization of $Y = \exp(XT)$ around X_k . The matrix exponential and its derivatives are delicate to evaluate numerically (Moler and Van Loan 2003, Al-Mohy and Higham 2009). Therefore, we use a method that relies on an off-the-shelf implementation of the matrix exponential.

PROPOSITION 7.2 *Consider $Y = \exp(XT) \in \mathbb{R}^{d \times d}$ and its linearization c_{k+1} around X_k . Let $g_{k+1} : \mathbb{R}^{d \times d} \mapsto \mathbb{R}^{d \times d}$ be the mapping with values*

$$g_{k+1}(E) = Y_{12},$$

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \exp \left(\begin{bmatrix} X_k & E \\ 0 & X_k \end{bmatrix} T \right) \in \mathbb{R}^{2d \times 2d}. \quad (31) \quad \text{define}$$

Then, $c_{k+1}(X) = \exp(X_k T) + \sum_{ij} (X_{ij} - X_{k,ij}) g_{k+1}(e_i e_j^\top)$.

When X is a transition rate matrix $X = Q$, one can use, instead, the linearization around Q_k

$$c_{k+1}(Q) = \exp(Q_k T) + \sum_{ij: i \neq j} (Q_{ij} - Q_{k,ij}) g_{k+1}(e_i e_j^\top - e_i e_i^\top). \quad (32)$$

8. Computational results

This section is designed with the goal of demonstrating the method developed and evaluating the extent to which information can be recovered from contract values. The discussion takes place in the context of point-to-point financial transmission rights.

8.1. FTR contract model

Consider an electricity transmission network with N nodes and L lines. The state of the system is defined by the load levels at nodes and the network topology. Hence, the system state space is $\mathcal{S} = \mathcal{S}^{\text{load}} \times \mathcal{S}^{\text{grid}}$, where $\mathcal{S}^{\text{load}}$ denotes the state space for the load, and $\mathcal{S}^{\text{grid}}$ denotes the state space of the network. Let π^S be the price formation mechanism returning the day-ahead hourly locational marginal prices (LMP) for all nodes at every state.

The instantaneous payoff from holding an obligation-type FTR k from node i_k to node j_k when the system is in state s^m is defined as

$$x_{k,m} = \pi_{j_k}^S(s^m) - \pi_{i_k}^S(s^m) \quad (33)$$

which denotes the difference of the day-ahead hourly locational marginal prices of nodes i_k and j_k in state s^m attributable to the congestion component of the price. The LMP for node i in state s^m is decomposed as a sum of three components: a system price component $\lambda^{\text{sys},m}$, a congestion component $\lambda_i^{\text{cong},m}$, and a loss component $\lambda_i^{\text{loss},m}$. The decomposition methodology varies among system operators, see (Eldridge *et al.* 2018). We

$$\pi_i^S(s^m) = \lambda^{\text{sys},m} + \lambda_i^{\text{cong},m}. \quad (34)$$

The system price $\lambda^{\text{sys},m}$ cancels out in (33) but is included to facilitate price comparisons. The FTR payoffs are based on the congestion component of the day-ahead locational marginal prices.

8.2. Description of the test system

We consider a power system model known as the PJM 5-node system (Li and Bo 2010), also available in Matpower (Zimmerman *et al.* 2011). figure 2(a) illustrates the system diagram. The model includes $N = 5$ nodes, labeled from A to E, $L = 6$ lines {AB, AD, AE, BC, CD, DE}, 5 generator units (indicated by double circles), and 3 loads at nodes B, C, D (indicated by gray circles).

The demand is assumed to realize two levels: High or Light. Hence, the state space for the load is $\mathcal{S}^{\text{load}} = \{d^{\text{light}}, d^{\text{high}}\}$. The load level at each node is determined by two variables, the active load in MW and the reactive load in Mvar. The demand states are described in figure 2(b). We assume that the transmission grid is in a state with at most one unplanned line outage. The state space of the grid is the set $\mathcal{S}^{\text{grid}} = \{u \in \{0,1\}^6 : \sum_{\ell=1}^6 (1 - u_\ell) \leq 1\}$ where u_ℓ is the 0-1 status of line ℓ . In total there are $n = 2 \times 7 = 14$ system states in the state space $\mathcal{S} = \mathcal{S}^{\text{load}} \times \mathcal{S}^{\text{grid}}$. We label the states from $m = 1$ to $m = 14$ as follows: $m = 1 + \ell^o + 7\delta$, where δ is the load indicator and ℓ^o is the label of the line which is out, or $\ell^o = 0$ if there is no outage.

At every system state, we compute the components of the price formation π^S in equation (34) to obtain the LMPs at every node using the Matpower software. Other input variables include capacity generator units in MW, bid cost data in \$/MW, and reactance x of each transmission line. The values of these input variables are specified in figure 2. The resistance for each line is $r = x/10$ in per unit. The lines AB and DE have a limited capacity of 400 MVA and 240 MVA, respectively; the other lines are modeled as unconstrained. Generator units are also able to produce or absorb reactive power in order to maintain the voltage magnitude at a mandated set point. The maximum reactive power produced is set to 0.75 times the MW capacity. The maximum reactive power absorbed is set to -0.75 times the MW capacity.

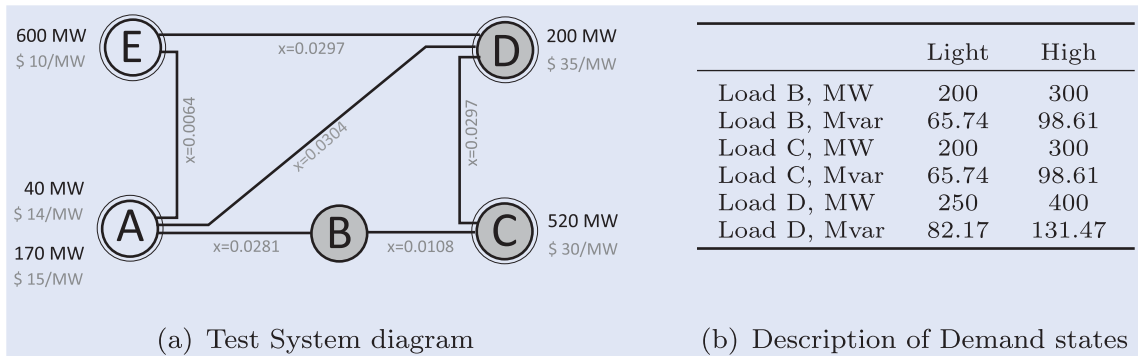


Figure 2. PJM 5-node Test System. (a) Test System diagram. (b) Description of Demand states.

8.3. Assumptions on the system dynamics

A simple model is adopted to describe the system dynamics. We let $S(t)$ follow a Markov chain (p_0, Q) where $Q \in \mathbb{Q}^{n \times n}$. Our assumptions are as follows.

- A1. The transitions between load states and between grid states cannot be simultaneous.
- A2. There is no direct transition from a state with $u_\ell = 0$ to another state with $u_{\ell'} = 0$, $\ell' \neq \ell$.

These assumptions imply a sparsity pattern for Q . To describe it, let $\bar{Q}^{\text{patt}}$ be a matrix where the non-diagonal elements of Q that can be nonzero are represented by ones. Under our state labeling convention, $\bar{Q}^{\text{patt}} = \begin{pmatrix} \Gamma & I \\ I & \Gamma \end{pmatrix}$, where I is the identity matrix of dimension 7, and $\Gamma \in \mathbb{R}^{7 \times 7}$ has elements $\Gamma_{ij} = 1$ if $i = 1$ and $j > 1$, $\Gamma_{ij} = 1$ if $j = 1$ and $i > 1$, and 0 otherwise.

8.4. Parametric study: implied probabilities from a single FTR contract

This section investigates the implied probabilities as a function of a given FTR market value y_{EC} for the annual contract from E to C. The annual FTR is defined over $\mathcal{T} = [0, T]$ with $T = 8760$. Given an annual discount rate r_y , the continuous-time discount rate is $r = \log(1 + r_y)/T$. We set $r_y = 2\%$. We seek to build a matrix Q consistent with the FTR market value y_{EC} . Then, we use the matrix Q to estimate the implied durations of the load levels, and the implied fractions of the year the capacitated lines AB and DE are congested. We also use the matrix Q to price other FTR annual contracts, namely, the contracts AB, EB, and ED.

We consider only 3 additional contracts because in a system with 5 nodes, at most 4 annual obligation-type FTRs can have their price set independently. This is formally expressed by the following proposition.

PROPOSITION 8.1 *Suppose the transmission grid has a single connected component with N nodes. Suppose all FTRs are obligations that are defined over the same time window. Then, a set of FTR prices contains at most $N - 1$ linearly independent FTR prices.*

Given that we have only one input and the estimation problem is underdetermined, we reduce the number of independent parameters for Q , using the following assumptions.

- A3. The lines have identical outage rates.
- A4. The lines have identical repair rates.
- A5. The outage rates and repair rates do not depend on the load level.
- A6. The demand state evolves independently from the grid state.

The initial distribution p_0 is assumed to be concentrated on the state $m = 1$. The payoff vector $x_{EC} \in \mathbb{R}^n$ has elements $x_{EC,m}$ defined by (33) with $i_k = E$, $j_k = C$. The explicit description of $x_{EC,m}$ is in appendix 2. The estimation problem is formulated as

$$\min \|Y_{12}x_{EC} - y_{EC}\|^2$$

$$\text{subject to } \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \exp \left(\begin{bmatrix} 0 & p_0 \\ 0_n & Q - rI \end{bmatrix} T \right),$$

$$Q1_n = 0, \quad Q_{ij} \geq 0 \text{ for } i \neq j,$$

$$Q_{ij} = 0 \text{ if } i \neq j \text{ and } \bar{Q}_{ij}^{\text{patt}} = 0, \quad [\text{A1--A2}]$$

$$Q_{1j} = Q_{12} = Q_{8,j+7} \text{ for } 2 \leq j \leq 7, \quad [\text{A3, A5}]$$

$$Q_{i1} = Q_{21} = Q_{i+7,8} \text{ for } 2 \leq i \leq 7, \quad [\text{A4, A5}]$$

$$Q_{i+7,i} = Q_{81},$$

$$Q_{i,i+7} = Q_{18} \text{ for } 2 \leq i \leq 7. \quad [\text{A6}] \quad (35)$$

The problem is solved by the successive linearization method described in Section 7. Let Q^* constitute an optimal solution of problem (35). We compute the row vector Y_{12}^* using the exact nonlinear equality constraint, where Y_{12}^* corresponds to the implied occupation measure $\mu_{T,r}$.

In figure 3, we present the results obtained as a function of the FTR market value y_{EC} of the FTR from E to C.

To obtain the implied load durations, we first calculate the undiscounted occupation measure using the first row of table 2 with $r = 0$, $T = 8760$ and $Q = Q^*$. Let $\mu_{T,0} \in \mathbb{R}^{1 \times n}$ denote the result. Given the state labels, the expected fraction of the year in high load is $v^{\text{high}} = \sum_{m=1}^7 \mu_{T,0,m}/T$. The expected fraction of the year in light load is $v^{\text{light}} = \sum_{m=8}^{14} \mu_{T,0,m}/T = 1 - v^{\text{high}}$. Figure 3 indicates that as the EC contract value y_{EC} increases, the expected duration of the high load level increases.

To obtain the implied line congestion durations, we use the AC power flow solutions to identify the set of states m where the flow of the line attains its MVA rating. Then we add up the undiscounted occupation measure of these states, and divide by T to get the fraction of the year the line is congested. We do this for the lines AB and ED which are the only lines in the test case subject to flow limits.

To price the other FTR contracts, we calculate the payoff vectors x_{AB} , x_{EB} and x_{ED} using (33). The details are in appendix 2. For $k = AB, EB, ED$, we set $v_k = Y_{12}^* x_k$. The FTR values are for 1 MW of right during one year.

The results for the sequence of values y_{EC} were generated by taking the solution of the previous problem as the initial point around which the matrix exponential constraint is linearized. This strategy accelerates the convergence to an optimal solution. Following a path along modified problems relates to continuation methods in nonlinear optimization. See (Nocedal and Wright 2006).

The dotted curve relative to the value of the FTR contract EC indicates that the model matches the target value. That is, the objective of (35) attains a zero optimal value. A lower bound for the global minimum is 0 by construction of the objective. Therefore, the local optimum to the nonconvex problem that was found is guaranteed to be a global optimum.

The optimal solution to (35) is not unique. We could have restricted the solution set further. The problem is degenerate if we set all outage probabilities to 0. In this case we have only 2 states. In the high-load state, the FTR payoff is $x_{EC}^{\text{high}} = 19.38$. See table A4. In the light-load state, the payoff is 0. The FTR value is then equal to $y_{EC} = \mu_{T,r}(T)v_{T,r}^{\text{high}}x_{EC}^{\text{high}}$ where $\mu_{T,r}(T) = (1 - e^{-rT})/r$. This gives the normalized *discounted* occupation of the high-load state,

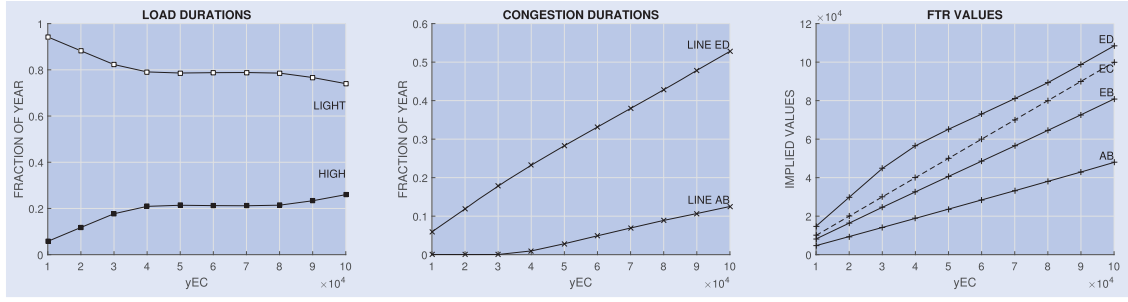


Figure 3. Implied load durations, line congestion durations and financial transmission right values, as a function of the target value y_{EC} of the FTR for EC.

$v_{T,r}^{\text{high}} = y_{EC} / (x_{EC}^{\text{high}} \mu_{T,r}(T))$, as a function of y_{EC} . By Proposition 6.5, the set of matrices Q compatible with the data is given by (24), where β in (24) is a parameter influencing the frequency of transition between the two load levels. The line ED is congested in the high-load state. The line AB is not congested in any of the 2 states. Thus the fraction of time the line ED is congested is $v_{T,0}^{\text{high}}$, calculated given Q , T and $r = 0$. The discrepancy between $v_{T,r}^{\text{high}}$ and $v_{T,0}^{\text{high}}$ is maximized with the smallest value of β but is quite small (r is itself close to 0). When y_{EC} varies as in figure 3, we find that $v_{T,0}^{\text{high}}$ varies almost linearly from $v_{T,0}^{\text{high}} = 0.06$ at $y_{EC} = 10^4$ to $v_{T,0}^{\text{high}} = 0.59$ at $y_{EC} = 10 \cdot 10^4$.

The values of the other FTRs implied by y_{EC} are $v_k = \mu_{T,r}(T) v_{T,r}^{\text{high}} x_k^{\text{high}} = (x_k^{\text{high}} / x_{EC}^{\text{high}}) y_{EC}$ for $k = AB, EB, ED$. From table A4 one finds $v_{AB} = 0.470 y_{EC}$, $v_{EB} = 0.819 y_{EC}$, $v_{ED} = 1.497 y_{EC}$. The implied values are independent of Q and r . The linear relationship comes from the fact that the FTRs have their nonzero payoff in the same high-load state.

8.5. Implied probabilities from multiple FTR contracts

In this section we consider the estimation of a matrix Q consistent with the values of the annual FTR contracts from E to C and from E to D. Let y_{EC}, y_{ED} be their values. Let $\epsilon = (\epsilon_1, \epsilon_2)$ be a vector of residuals. We minimize $\|\epsilon\|^2$

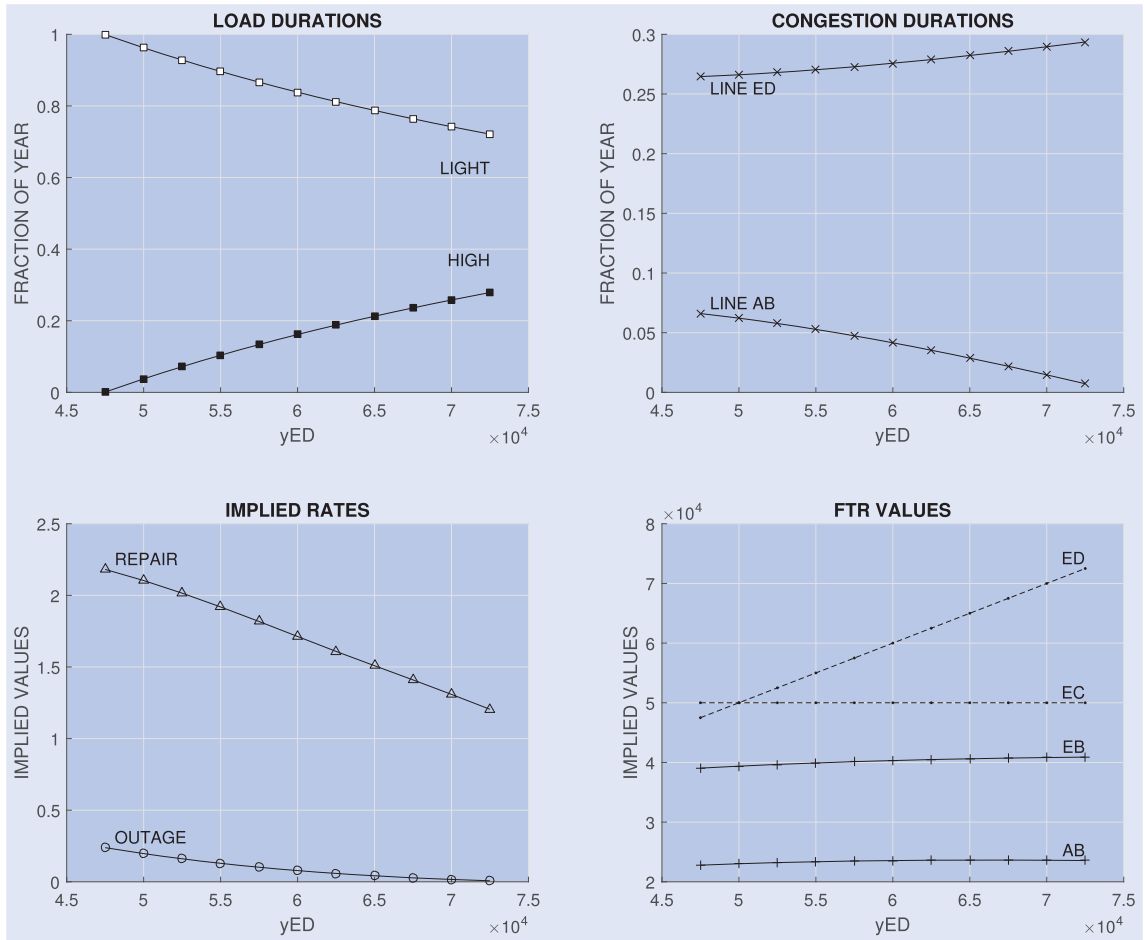


Figure 4. Implied load durations, line congestion durations, outage rates, and financial transmission right values as a function of the FTR value y_{ED} for ED.

subject to $y_{EC} = Y_{12}x_{EC} + \epsilon_1$, $y_{ED} = Y_{12}x_{ED} + \epsilon_2$, $\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \exp\left(\begin{bmatrix} 0 & p_0 \\ 0_n & Q - rI \end{bmatrix} T\right)$, and subject to the constraints on Q as in (35). We present the results as we vary the value y_{ED} for the FTR for ED, with the FTR value y_{EC} set to $5 \cdot 10^4$. The results are depicted in figure 4. This time, setting two FTR values independently allows us to start distinguishing between two factors of congestion in the system: the duration of high load, versus the frequency and duration of the unplanned outages. Further numerical experiments have shown to us that under our structural assumptions on Q , and for our test case that has 2 load levels and 2 capacitated transmission elements, it did not seem to be possible or useful to set the value of 3 inputs independently.

9. Conclusions

This paper considers derivatives for which the price of the underlying comes from a structural model and the payoffs can be received over a continuous or discrete period of time. The structural model is driven by a set of Markovian inputs which can be computationally intensive to evaluate. This paper proposes an approach to compute the implied transition probabilities of the stochastic input processes of the structural model from the derivative's available market data.

The method relies on a decomposition of the contract value into an expected discounted occupation measure component and a payoff component. The occupation measure component captures the information on the stochastic dynamics of the inputs, the discount rate, and the payoff times. The evaluation of the two components is decoupled, which brings considerable computational benefits. Next, we establish explicit expressions for the occupation measures and implied transition functions, which are instrumental to compute implied transition probabilities from implied occupation measures. We show that the calibration of the model can then be formulated as a structured regression problem with the occupation measure is nonlinear constraint. Finally, we develop a numerical solution method to solve this optimization problem.

We present an illustrative application focused on financial transmission rights from electricity markets. These contracts pay the difference between electricity prices at two nodes for a specified future time period. The price formation mechanism includes a large-scale optimization scheme to maximize the social-welfare subject to network capacity constraints with inputs such as network structure and nodal demands. Extensive parametric study of the computed implied probabilities of network congestion implied from the observation of these contracts is carried out for a standard benchmark network.

Beyond our illustrative application, note that other types of constraints on the transition matrices could be accommodated. For instance, if the states of the Markov chain result from the quantization of a continuous state space, the transition or rate matrices to be calibrated may depend on fewer parameters. In this case, the regression problem can have additional constraints to express the relation between the parameters and the transition matrices. This regularization device will structure the transition matrices and occupation measures implied from market data.

While we have described our methodology using stationary Markov chains, non-stationary and periodic Markov processes can be cast in our framework by associating a time or regime index to the state. This results in large but sparse transition matrices. For non-Markovian processes, scenario trees could serve as a simple finite representation over a finite horizon. However, as the optimization techniques to fit occupation measures on scenario trees will be different, this could be the subject of subsequent studies. In general, with arbitrary stochastic processes defined by a limited number of parameters to calibrate, the occupation measures of our decomposition can be evaluated numerically by Monte Carlo simulation.

Developing more efficient optimization methods when the Markov process includes a large number of states remains an interesting direction of future work. Evaluating the efficiency of our methodology on other derivatives such as credit default swaps could be another fruitful direction for future research.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Appendix 1: Proofs of statements

Proof of Proposition 4.1. Starting from (3) and partitioning $\mathcal{T}$ as $\mathcal{T} = \biguplus_{m=1}^n \mathcal{T}_m$, we have in continuous-time

$$\begin{aligned} V_{f,\mathcal{T}} &= \mathbb{E} \left[\int_{t \in \mathcal{T}} e^{-rt} f(\pi^S(S(t))) dt \right] \\ &= \mathbb{E} \left[\sum_{m=1}^n \int_{\mathcal{T}_m} e^{-rt} f(\pi^S(s^m)) dt \right] \\ &= \sum_{m=1}^n f(\pi^S(s^m)) \mathbb{E} \left[\int_{\mathcal{T}_m} e^{-rt} dt \right] = \sum_{m=1}^n f(\pi^S(s^m)) \mu(\mathcal{T}_m). \end{aligned}$$

From there we use (8), as described in (9). The reformulation is similar in discrete-time. ■

Proof of Proposition 6.1. We start by recalling useful facts about the transition rate matrix Q . The matrix Q satisfies $Q1_n = 0_n$ and $Q_{ij} \geq 0$ for $i \neq j$. Therefore, Q is singular but $(rI - Q)$ with $r > 0$ is invertible. To see this, note that $Q1_n = 0_n$ shows that Q admits the eigenvalue 0 and is thus singular. Furthermore, as Q is row-diagonally dominant with $Q_{ii} \leq 0$, Gershgorin's circle theorem implies that each eigenvalue of Q has a nonpositive real part. It follows that for any $r > 0$, the matrix $(rI - Q)$ has all its eigenvalues with positive real parts, thus in particular it is invertible.

In continuous time, the vector of state probabilities at time t is given by $p(t) = p_0 \exp(Qt)$, using the matrix exponential function, defined by $\exp(X) = \sum_{k=0}^{\infty} (X^k/k!)$.

- For $\mathcal{T} = \mathbb{R}_T$ and $r \geq 0$, $\mu = \int_0^T e^{-rt} p(t) dt = \int_0^T e^{-rt} p_0 \exp(Qt) dt = \int_0^T p_0 \exp((Q - rI)t) dt$. Theorem 1 in Van Loan (1978) shows that for any matrices $A \in \mathbb{R}^{n_1 \times n_1}$, $B \in \mathbb{R}^{n_1 \times n_2}$, $C \in \mathbb{R}^{n_2 \times n_2}$,

$$\begin{aligned} &\exp \left(\begin{bmatrix} A & B \\ 0 & C \end{bmatrix} T \right) \\ &= \begin{bmatrix} \exp(AT) & \int_0^T \exp(A(T-t))B \exp(Ct) dt \\ 0 & \exp(CT) \end{bmatrix}. \end{aligned}$$

Setting $A = 0$, $B = p_0$, $C = Q - rI$ we get $\mu = Y_{12}$ where $\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \exp \left(\begin{bmatrix} 0 & p_0 \\ 0_n & Q - rI \end{bmatrix} T \right)$. This establishes row 1 of table 2.

Alternatively, for a nonsingular matrix D , we have $\int_0^T \exp(-Dt) dt = (I - \exp(-DT))D^{-1}$. We set $D = rI - Q$. This establishes row 2 of table 2.

- For $\mathcal{T} = \mathbb{R}_+$ and $r > 0$, $\mu = p_0 \int_0^{\infty} \exp((-rI - Q)t) dt = p_0(rI - Q)^{-1}$, which establishes row 3 of table 2. In discrete time, the vector of state probabilities at time t is given by $p(t) = p_0 P^t$, using the matrix power function. The matrix P satisfies $P1_n = 1_n$, thus P admits 1 as an eigenvalue, implying that $I - P$ is singular. However $I - \gamma P$ with $0 < \gamma < 1$ is invertible.
- For $\mathcal{T} = \mathbb{N}_{T-1}$ and $\gamma = 1$, $\mu = \sum_{t=0}^{T-1} p(t) = p_0 \sum_{t=0}^{T-1} P^t$. We use induction on $T \geq 1$ to show that $\begin{bmatrix} 1 & p_0 \\ 0_n & P \end{bmatrix}^T = \begin{bmatrix} 1 & p_0 \sum_{t=0}^{T-1} P^t \\ 0 & P^T \end{bmatrix}$, which justifies row 4 of table 2.
- For $\mathcal{T} = \mathbb{N}_{T-1}$ and $0 < \gamma < 1$, we have $\mu = \sum_{t=0}^{T-1} \gamma^t p(t) = p_0 \sum_{t=0}^{T-1} (\gamma P)^t$. As $I - \gamma P$ is invertible, we can directly evaluate $\sum_{t=0}^{T-1} (\gamma P)^t = (I - \gamma^T P^T)(I - \gamma P)^{-1}$, which justifies row 5 of table 2.

- For $\mathcal{T} = \mathbb{N}$ and $0 < \gamma < 1$, $\mu = p_0 \sum_{t=0}^{\infty} (\gamma P)^t$ and $\sum_{t=0}^{\infty} (\gamma P)^t = (I - \gamma P)^{-1}$, which justifies row 6 of table 2. ■

Proof of Proposition 6.2. Equations (16)–(19) reformulate the results of table 2, except that p_0, v are now fixed quantities, while P and Q are matrix variables. The matrix inverses seen in table 2 are cancelled out by right-multiplication on both sides of the equalities. ■

Proof of Proposition 6.3, Infinite-Horizon. First, it is noted that $p_{0,k}/\mu(T) = p_{0,k}(1 - \gamma)$ is a lower bound for v_k , achieved when the Markov chain does not visit state k after time 0, using any P such that $Pe_k = 0_n$. Also, $[p_{0,k} + \gamma/(1 - \gamma)](1 - \gamma)$ is an upper bound for v_k , achieved when the Markov chain remains in state k after time 0, using $P = 1_n e_k^\top$. This shows that the domain is included in the box

$$B_\gamma(p_0) := [(1 - \gamma)p_0, (1 - \gamma)p_0 + \gamma 1_n^\top].$$

Now, let v be any row vector in $\mathcal{P}^n \cap B_\gamma(p_0)$. Define a row vector $d_v^\top$ and a rank-1 matrix P_v as

$$d_v^\top = \gamma^{-1}(v - (1 - \gamma)p_0), \quad P_v = 1_n d_v^\top.$$

Observe that $d_v^\top 1_n = 1$ (and thus $P_v 1_n = 1_n$). Observe that $d_v \geq 0$ (and thus $P_v \geq 0$) since $v \in B_\gamma(p_0)$. Hence $P_v \in \mathcal{P}^{n \times n}$. Finally, observe that $P_v \in \mathcal{C}_{\mathbb{N}, \gamma}^{-1}(p_0, v)$ since P_v also verifies $(1 - \gamma)p_0 + \gamma v P_v = v$. The existence of P_v shows that $\mathcal{C}_{\mathbb{N}, \gamma}^{-1}$ is nonempty, thus v is in $\text{dom } \mathcal{C}_{\mathbb{N}, \gamma}^{-1}(p_0, \cdot)$, which can now be identified as $\mathcal{P}^n \times B_\gamma(p_0)$. The domain is convex, as the intersection of convex sets. ■

Proof of Proposition 6.3, Finite Horizon. Consider $\mathcal{T} = \mathbb{N}_{T-1}$. Recall from table 1 that $\mu(T) = \sum_{t \in \mathcal{T}} \gamma^t = \frac{1 - \gamma^T}{1 - \gamma}$. Note that $p_{0,k}/\mu(T) = p_{0,k} \frac{1 - \gamma}{1 - \gamma^T}$ is a lower bound for v_k , achieved when the Markov chain does not visit state k after time 0, using any P such that $Pe_k = 0_n$. Also, $[p_{0,k} + \frac{\gamma - \gamma^T}{1 - \gamma}]/\mu(T) = p_{0,k} \frac{1 - \gamma}{1 - \gamma^T} + \frac{\gamma - \gamma^T}{1 - \gamma^T}$ is an upper bound for v_k , achieved when the Markov chain remains in state k after time 0, using $P = 1_n e_k^\top$. This shows that the domain is included in the box

$$B^{\gamma, p_0} := \left[\left(\frac{1 - \gamma}{1 - \gamma^T} \right) p_0, \left(\frac{1 - \gamma}{1 - \gamma^T} \right) p_0 + \left(\frac{\gamma - \gamma^T}{1 - \gamma^T} \right) 1_n^\top \right].$$

Now, let v be any row vector in $\mathcal{P}^n \cap B^{\gamma, p_0}$. Define a row vector $d_v^\top$ and a rank-1 matrix P_v as

$$d_v^\top := \left(\frac{1 - \gamma^T}{\gamma - \gamma^T} \right) v - \left(\frac{1 - \gamma}{\gamma - \gamma^T} \right) p_0, \\ P_v := 1_n d_v^\top.$$

Observe that $d_v^\top 1_n = 1$ (and thus $P_v 1_n = 1_n$). Observe that $d_v \geq 0$ (and thus $P_v \geq 0$) since $v \in B^{\gamma, p_0}$. Hence $P_v \in \mathcal{P}^{n \times n}$. Finally, observe that $P_v \in \mathcal{C}_{\mathbb{N}_{T-1}, \gamma}^{-1}(p_0, v)$ since P_v also verifies $v(I - \gamma P_v) = (\frac{1 - \gamma}{1 - \gamma^T}) p_0 (I - \gamma^T P_v^\top)$. To see this, note that for any rank-one matrix $P = ab^\top$ and integer $k \geq 2$, we have $P^k = ab^\top ab^\top P^{k-2} = (b^\top a) ab^\top P^{k-2} = (a^\top b) P^{k-1}$ and thus by induction $P^k = (a^\top b)^{k-1} P$; the formula remains valid for $k = 1$. Since $P_v = 1_n d_v^\top$ and $d_v^\top 1_n = 1$, we get $P_v^T = P_v$ for $T \geq 1$.

The existence of P_v shows that $\mathcal{C}_{\mathbb{N}_{T-1}, \gamma}^{-1}$ is nonempty, thus v is in the $\text{dom } \mathcal{C}_{\mathbb{N}_{T-1}, \gamma}^{-1}(p_0, \cdot)$, which can be identified as $\mathcal{P}^n \times B^{\gamma, p_0}$. This proves (22).

The domain is convex, as the intersection of convex sets. We now show that it is increasing in T . From (22), the lower bound of v_i is $p_{0,i}(1 - \gamma)/(1 - \gamma^T)$ which is decreasing in T . The upper

bound of v_i is $p_{0,i} \frac{1 - \gamma}{1 - \gamma^T} + 1 \frac{\gamma - \gamma^T}{1 - \gamma^T} = p_{0,i} + (1 - p_{0,i}) \frac{\gamma - \gamma^T}{1 - \gamma^T}$ which is increasing in T . This shows that $T < T'$ implies $\text{dom } \mathcal{C}_{\mathbb{N}_{T-1}, \gamma}^{-1}(p_0, \cdot) \subset \text{dom } \mathcal{C}_{\mathbb{N}_{T'-1}, \gamma}^{-1}(p_0, \cdot)$. Finally, the set-valued mapping is continuous and converges to (21) as $T \rightarrow \infty$, showing (21). A direct proof of (21) is possible as well. ■

Proof of Proposition 6.3, Case with Additional Constraints. The domain of $v \mapsto \mathcal{C}^{-1}(p_0, v) \cap \mathcal{X}$ can only be a subset of the domain of $v \mapsto \mathcal{C}^{-1}(p_0, v)$, since new constraints can only eliminate some transition matrices and therefore cannot add new values of v to the domain.

To establish the claim that the domain of $v \mapsto \mathcal{C}^{-1}(p_0, v) \cap \mathcal{X}$ is not necessarily convex, a counterexample suffices. For instance, consider a 3-state Markov chain (p_0, P) such that $p_0 = [0 \ 0 \ 1]$. Furthermore, restrict the set of possible transition matrices to $\mathcal{X} = \{P \in \mathcal{P}^{3 \times 3} : P_{31} = 0\}$.

Consider the case $T = 3$, $\gamma = 1$, so the occupation measure is $v = p_0(I + P + P^2)/3$. Observe that $P^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ realizes $v^{(1)} = [1 \ 1 \ 1]/3$ and $P^{(2)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ realizes $v^{(2)} = [0 \ 0 \ 1]$. Thus, $v^{(1)}, v^{(2)} \in \text{dom } \mathcal{C}_{\mathbb{N}_{T-1}, \gamma}^{-1}(p_0, \cdot)$ since $P_{31}^{(1)} = P_{31}^{(2)} = 0$. However, $v' = (v^{(1)} + v^{(2)})/2 = [1 \ 1 \ 4]/6$ is not in the domain, as no P with $P_{31} = 0$ can realize v' , establishing the nonconvexity claim.

To prove that no P exists that can attain v' , we write the condition $p_0(I + P + P^2)/3 = v'$ with $P_{31} = 0$. This gives two independent equations, $P_{32}P_{21} = 3v'_1 = 1/2$ and $P_{32} + (1 - P_{21} - P_{23})P_{32} + (1 - P_{32})P_{32} = 3v'_2 = 1/2$. Multiplying the second equation by $2P_{21}$ gives $1 + (1 - P_{21} - P_{23}) + (1 - P_{32}) = 2P_{21}$, that is, $2P_{21} + P_{23} + P_{32} = 3$. The only possible solution with $P_{32} \leq 1$, $P_{23} \geq 0$, and $P_{22} = 1 - P_{21} - P_{23} \geq 0$ is $P_{23} = 0$, $P_{32} = 1$, $P_{21} = 1$. However, this contradicts $P_{32}P_{21} = 1/2$, so the solution set is empty. ■

The particular discrete-time solutions in Proposition 6.4 are established in the course of the proof of Proposition 6.3, thus below only the part on the continuous case is treated.

For the time set $\mathbb{R}_T$ with $r = 0$, a solution $Q_v \in \mathcal{C}_{\mathbb{R}_T, 0}^{-1}(p_0, v)$ is given by

$$Q_v = \beta(1_n b_v^\top - I), \quad b_v^\top = \left(\frac{\beta T v - (1 - e^{-\beta T}) p_0}{\beta T - (1 - e^{-\beta T})} \right), \\ \beta \geq \frac{c + W(-ce^{-c})}{T}, \quad c = c_0 = \max_m \left\{ \frac{p_{0,m}}{v_m} \right\}.$$

Proof of Proposition 6.4, Continuous-Time, Infinite-Horizon. Suppose heuristically, for column vectors $a, b \geq 0$, that

$$Q_v = ab^\top - (b^\top 1_n) \text{diag}(a),$$

which guarantees by construction that $Q_v \in \mathcal{Q}^{n \times n}$. Furthermore, with the ansatz $a = 1_n$, we have $Q_v = 1_n b^\top - (b^\top 1_n) I$.

Substituting Q_v into $rp_0 + vQ = rv$ (obtained from (17)), where we recall that p_0, v are row vectors, we get $rp_0 + b^\top - v(b^\top 1_n) = rv$.

A solution b , if it exists, satisfies $b^\top 1_n = \beta$ for some $\beta \geq 0$. This would give $b^\top = (\beta + r)v - rp_0$. It can be checked that $b^\top 1_n = \beta + r - r = \beta$, which leaves β as a degree of freedom to get a feasible solution for b . The last condition is $b^\top \geq 0$, hence we can choose any

$$\beta \geq \min\{\beta_0 \geq 0 : \beta_0 v \geq r(p_0 - v)\} = r \max_m \{ \left[\frac{p_{0,m}}{v_m} - 1 \right]^+ \} \\ = r \max_m \{ \frac{p_{0,m}}{v_m} - 1 \},$$

noting that there is always some m such that $\frac{p_{0,m}}{v_m} \geq 1$. We adopt the convention $0/0 = 0$ for the ratios.

Setting $c_0 = \max_m \{p_{0,m}/v_m\}$, we get the condition $\beta + r \geq rc_0$. With the changes of variable $\beta' = \beta + r$ and $b' = b/\beta$ we

obtain $\beta' \geq rc_0$, $[b']^\top = (\beta'v - rp_0)/(\beta' - r)$, and $Q_v = 1_n(\beta' - r)[b']^\top - (\beta' - r)I$. The result is stated with the primes removed from the entities. ■

Proof of Proposition 6.4, Continuous-Time, Finite-Horizon. Suppose heuristically, for column vectors $a, b \geq 0$,

$$Q_v = ab^\top - (b^\top 1_n)\text{diag}(a),$$

which guarantees by construction that $Q_v \in \mathcal{Q}^{n \times n}$. Furthermore, with the ansatz $a = 1_n$, we have $Q_v = 1_n b^\top - (b^\top 1_n)I$. If a feasible solution $b^\top$ exists, we can define $\beta = b^\top 1_n$. We have

$$\begin{aligned} Q_v^2 &= (1_n b^\top - (b^\top 1_n)I)(1_n b^\top - (b^\top 1_n)I) \\ &= -(b^\top 1_n)1_n b^\top + (b^\top 1_n)^2 I = -\beta Q_v, \end{aligned}$$

and by induction, $Q_v^k = (-\beta)^{k-1} Q_v$ for $k \geq 1$, from which we obtain

$$\begin{aligned} \exp(Q_v T) &= I + \sum_{k=1}^{\infty} \frac{(-\beta)^{k-1} T^k}{k!} \\ Q_v &= I + Q_v \frac{1}{-\beta} \left[-1 + \sum_{k=0}^{\infty} \frac{(-\beta T)^k}{k!} \right] \\ &= I + \frac{1 - e^{-\beta T}}{\beta} Q_v = e^{-\beta T} I + \frac{1 - e^{-\beta T}}{\beta} 1_n b^\top. \end{aligned}$$

Substituting into $rp_0(I - \exp(-(rI - Q)T))/(1 - e^{-rT}) + vQ = rv$ (equation (17)), we get

$$rp_0 \left(\frac{I - e^{-rT} [e^{-\beta T} I + \frac{1 - e^{-\beta T}}{\beta} 1_n b^\top]}{1 - e^{-rT}} \right) + (b^\top - \beta v) = rv,$$

which leads, after some algebra, to

$$\begin{aligned} b^\top &= \beta \left(\frac{(r + \beta)(1 - e^{-rT})v - r(1 - e^{-(r+\beta)T})p_0}{-re^{-rT}(1 - e^{-\beta T}) + \beta(1 - e^{-rT})} \right) \\ &= \beta \left(\frac{(r + \beta)(1 - e^{-rT})v - r(1 - e^{-(r+\beta)T})p_0}{(r + \beta)(1 - e^{-rT}) - r(1 - e^{-(r+\beta)T})} \right). \end{aligned}$$

It is easy to check $b^\top 1_n = \beta$, which leaves β as a degree of freedom to ensure $b^\top \geq 0$. The condition for the denominator to be nonnegative is

$$\frac{(r + \beta)T}{1 - e^{-(r+\beta)T}} \geq \frac{rT}{1 - e^{-rT}},$$

which is always verified since the function $x \mapsto x/(1 - e^{-x})$ is increasing in $x \geq 0$. The condition for the numerator to be nonnegative is

$$\frac{(r + \beta)T}{1 - e^{-(r+\beta)T}} \geq \frac{rT}{(1 - e^{-rT})} \max_m \left\{ \frac{p_{0,m}}{v_m} \right\} := c.$$

Setting $x = (r + \beta)T$, we need $x/(1 - e^{-x}) \geq c$, that is, $x + ce^{-x} - c \geq 0$. The tightest solution is given by $x_0 = c + W(-ce^{-c})$, where W denotes the (upper branch of the) Lambert-W function (see e.g. Corless *et al.* 1996). As $c \geq 1$, we have $-ce^{-c} \in [-e^{-1}, 0)$, thus $-ce^{-c}$ is in the domain of W , with $W(-ce^{-c}) \in [-1, 0]$. With the change of variable $\beta' = r + \beta$, we get the condition $\beta' \geq x_0/T$.

Finally, with the change of variable $b' = b/\beta = b/(\beta' - r)$, we get the stated solution, where we have renamed β' to β and b' to b_v .

If we select the minimal value $\beta' = x_0/T$, then $\beta'^{-1}(1 - e^{-rT}) = (1 - e^{-\beta'T})c_0$ with $c_0 = \max_m \{p_{0,m}/v_m\}$, which implies that $(b')^\top = b_v^\top = (c_0 v - p_0)/(c_0 - 1)$, independent of r or T . ■

Proof of Proposition 6.5. In dimension 2, $Q \in \mathcal{Q}^{2 \times 2}$ is of rank at most 1, thus the particular solutions of rank 1 cover all possible cases. ■

Proof of Proposition 7.1. By induction on T , one can check that $Y_{11} = Y_{22} = X_k^T$, and $Y_{12} = \sum_{\ell=0}^{T-1} X_k^\ell EX_k^{T-1-\ell}$. By decomposing $X - X_k$ as $X - X_k = \sum_{ij} (X_{ij} - X_{ij,k}) e_i e_j^\top$, we have

$$\begin{aligned} f_{k+1}(X) &= X_k^T + \sum_{\ell=0}^{T-1} X_k^\ell \left[\sum_{ij} (X_{ij} - X_{ij,k}) e_i e_j^\top \right] X_k^{T-1-\ell} \\ &= X_k^T + \sum_{ij} (X_{ij} - X_{ij,k}) \sum_{\ell=0}^{T-1} P_k^\ell (e_i e_j^\top) P_k^{T-1-\ell} \\ &= X_k^T + \sum_{ij} (X_{ij} - X_{ij,k}) g_k(e_i e_j^\top). \end{aligned}$$

Proof of Proposition 7.2. The directional derivative of $\exp(XT)$ at X_k in the direction E is equal to

$$F_E[\exp(XT)](X_k) = \int_0^T [\exp(X_k(T-t))E\exp(X_k t)] dt.$$

Let g_{k+1} be defined as in the proposition. We use the property, proved by Van Loan (1978), that $F_E[\exp(XT)](X_k) = g_{k+1}(E)$, and then the decomposition $X - X_k = \sum_{ij} (X_{ij} - X_{ij,k}) (e_i e_j^\top)$. When $X = Q$, the relations $Q1_n = 0_n$ and $Q_k 1_n = 0_n$ imply that we can omit the diagonal terms in the sum if we replace the directions $e_i e_j^\top$ by the feasible directions $(e_i e_j^\top - e_i e_i^\top)$. ■

Proof of Proposition 8.1. For each node i of the network, let $\lambda_i^{\text{cong}}(t)$ denote the congestion component of its day-ahead locational marginal price at time t . Define the expected discounted cumulated congestion price of i as

$$\varepsilon_i^{\text{cong}} := \mathbb{E} \left[\int_T e^{-rt} \lambda_i^{\text{cong}}(t) dt \right].$$

Now, let MST be a minimum spanning tree for the network. For each edge ij of MST, an FTR can be defined with a value per MW equal to

$$v_{ij} = \mathbb{E} \left[\int_T e^{-rt} [\lambda_j^{\text{cong}}(t) - \lambda_i^{\text{cong}}(t)] dt \right] = \varepsilon_j^{\text{cong}} - \varepsilon_i^{\text{cong}}.$$

For a FTR from any i' to any j' , let $i' = i_1, i_2, \dots, i_{n'} = j'$ describe the unique path between i' and j' along the MST. Then

$$v_{i'j'} = \varepsilon_{j'}^{\text{cong}} - \varepsilon_{i'}^{\text{cong}} = \sum_{k=1}^{n'-1} (\varepsilon_{i_{k+1}}^{\text{cong}} - \varepsilon_{i_k}^{\text{cong}}) = \sum_{k=1}^{n'-1} v_{i_k i_{k+1}}.$$

This indicates that the FTR price is a linear combination of the subset of FTR prices defined for the edges of the MST. Since an MST on a graph of N nodes has $N - 1$ edges, the result follows. ■

Appendix 2: Supplementary simulation details

This section reports on the locational marginal prices that are needed to construct the payoff vectors used in Sections 8.4 and 8.5. Following (33), for each state s^m , we have $x_{EC,m} = \pi_C^S(s^m) - \pi_E^S(s^m)$, where $\pi_i^S(s^m) = \lambda^{\text{syst},m} + \lambda_i^{\text{cong},m}$ by (34). We report the values $\pi_i^S(s^m)$ for each node i and each state s^m , comparing three methods of calculation.

Table A1: $\pi_i^S(s^m)$ is approximated by taking $\lambda_i^{\text{dc},m}$ as the local marginal price (LMP) at node i of a lossless DC power flow. The DC power flow calculates the branch flows from the active power nodal injections using the linear model $f^{\text{lines},m} = M^{\text{PTDF},m} p^{\text{inj},m}$. The matrix $M^{\text{PTDF},m} \in \mathbb{R}^{L \times N}$ is the matrix of power transfer distribution factors (see Kirschen and Strbac 2018) in state s^m .

Table A2: $\pi_i^S(s^m)$ is approximated by taking $\lambda_i^{\text{ac},m}$ as the LMP at node i of an AC power flow. The LMP is identified as the Lagrange multiplier of the real power balance constraint at node i .

Table A3: $\pi_i^S(s^m)$ is $\pi_i^S(s^m) = \lambda^{\text{syst},m} + \lambda_i^{\text{cong},m}$ where $\lambda_i^{\text{cong},m} = (M^{\text{PTDF},m})^\top (v_i^m - v_j^m)$, $M^{\text{PTDF},m} \in \mathbb{R}^{L \times N}$ is the matrix of power

Table A1. Prices $\lambda_i^{\text{dc},m}$.

	No outage	Outage of a line						
		AB	AD	AE	BC	CD	DE	
LMP at A	16.98	14.35	12.83	35.00	15.00	15.00	15.00	} High load
B	26.38	35.00	25.23	35.00	15.00	30.00	32.70	
C	30.00	35.00	30.00	35.00	30.00	30.00	30.00	
D	39.94	35.00	43.11	35.00	30.00	35.00	22.59	
E	10.00	10.00	10.00	10.00	11.84	10.79	15.00	
LMP at A	15.00	13.48	12.13	30.00	13.48	15.00	15.00	} Light load
B	15.00	30.00	21.50	30.00	13.48	15.00	15.00	
C	15.00	30.00	25.10	30.00	30.00	15.00	15.00	
D	15.00	30.00	35.00	30.00	30.00	15.00	15.00	
E	15.00	10.00	10.00	10.00	10.00	15.00	15.00	

Table A2. Prices $\lambda_i^{\text{ac},m}$.

	No outage	Outage of a line						
		AB	AD	AE	BC	CD	DE	
LMP at A	16.85	14.32	12.89	34.84	15.00	15.00	15.00	} High load
B	26.52	35.34	25.43	35.04	15.25	29.95	32.55	
C	30.00	35.14	30.00	34.93	30.00	30.00	30.00	
D	39.28	35.00	42.82	35.00	30.00	35.00	22.87	
E	10.00	10.00	10.00	10.00	11.82	10.78	14.91	
LMP at A	15.00	13.41	12.23	29.82	14.00	15.00	15.00	} Light load
B	15.19	30.11	21.84	30.03	14.15	29.89	15.25	
C	15.21	30.00	25.35	30.00	30.00	30.00	15.28	
D	15.11	29.73	35.00	29.92	29.78	15.06	15.24	
E	14.94	10.00	10.00	10.00	10.66	14.93	14.90	

Table A3. Prices $\pi^S(m) = \lambda^{\text{sys},m} + \lambda_i^{\text{cong},m}$.

	No outage	Outage of a line						
		AB	AD	AE	BC	CD	DE	
LMP at A	17.03	14.63	12.97	35.00	15.26	15.27	15.58	} High load
B	26.14	35.00	25.20	35.00	15.26	29.64	32.58	
C	29.65	35.00	29.90	35.00	30.00	29.64	29.99	
D	39.28	35.00	42.82	35.00	30.00	35.00	22.87	
E	10.27	10.34	10.19	10.41	12.16	11.12	15.58	
LMP at A	15.11	13.67	12.23	29.92	14.25	15.06	15.24	} Light load
B	15.11	29.73	21.56	29.92	14.25	29.38	15.24	
C	15.11	29.73	25.14	29.92	29.78	29.38	15.24	
D	15.11	29.73	35.00	29.92	29.78	15.06	15.24	
E	15.11	10.29	10.11	10.35	10.98	15.06	15.24	

transfer distribution factors in state s^m , and the nonnegative vectors $v_l^m, v_f^m \in \mathbb{R}^L$ are the vectors of dual variables associated to the line capacity constraints in state s^m which are of the form $-F^{\text{max},m} \leq f^{\text{lines},m} \leq F^{\text{max},m}$.

Since node D is the reference node, $\lambda^{\text{sys},m} = \lambda_D^{\text{ac},m}$.

Table A4: This table reports the hourly payoffs $x_{AB}, x_{EB}, x_{EC}, x_{ED}$ per MW of financial transmission rights from A to B, E to B, E to C, E to D, based on the differences of prices from table A3.

Table A4. Hourly payoffs $x_{k,m}$ per MW of Financial Transmission Right.

	No outage	Outage of a line						
		AB	AD	AE	BC	CD	DE	
FTR A→B	9.11	20.37	12.23	0.00	0.00	14.37	17.00	} High load
E→B	15.87	24.66	15.01	24.59	3.10	18.52	17.00	
E→C	19.38	24.66	19.71	24.59	17.84	18.52	14.41	
E→D	29.01	24.66	32.63	24.59	17.84	23.88	7.29	
FTR A→B	0.00	16.06	9.33	0.00	0.00	14.32	0.00	} Light load
E→B	0.00	19.44	11.45	19.57	3.27	14.32	0.00	
E→C	0.00	19.44	15.03	19.57	18.80	14.32	0.00	
E→D	0.00	19.44	24.89	19.57	18.80	0.00	0.00	